COHERENT QUADRANT

Battery Fire Safety Industry Analysis

Global

115 Manufacturer scored on Product Capability (X) and Business Capability (Y)

Market Classification

Market

Battery Fire Safety Industry Analysis

Market Type

The industry focused on developing, manufacturing, and implementing thermal management, containment, and suppression systems to prevent or mitigate thermal runaway in lithium-ion and other advanced batteries.

Provider Categories in this Market

1. Manufacturer

I chose Manufacturer because the core of the battery fire safety industry revolves around physical protection materials, coatings, and containment hardware that companies produce under their own names to physically isolate thermal events.

115	20	20	4
Companies assessed	Companies on the Coherent Quadrant	Scoring parameters	Quadrants

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Market Scoring Parameters

X — Product Capability

1. **Fire Suppression Technology & Performance**
2. **Battery Chemistry Compatibility & Coverage**
3. **Product Portfolio Breadth & Modularity**
4. **R&D and Innovation Pipeline**
5. **Certifications & Standards Compliance**

Y — Business Capability

1. **Named Customers & Contracts**
2. **Revenue & Growth**
3. **Geographic Footprint & Channel Network**
4. **Installed Base & Market Share**
5. **Funding & Strategic Partnerships**

Quadrant Positioning

(20 on chart · 115 in total)



Countries on chart (by quadrant)

Leaders: United States×2, United Kingdom×1, Canada×2 · **Challengers**: Netherlands×1, Germany×1, United Kingdom×1, United States×1, South Korea×1 · **Trailblazers**: United States×2, Norway×1, Austria×1, Germany×1 · **Evolving Players**: Netherlands×1, South Korea×1, Australia×1, UAE×1, India×1

What Each Quadrant Means

Leaders **Integrated Market Leaders**

Best-in-class Product Capability, Best-in-class Business Capability

Companies demonstrating strong product capabilities and robust business execution. They combine mature and differentiated offerings with established market presence, customer reach, commercial scale, and effective growth strategies, positioning them strongly for sustained market leadership.

Challengers **Scale-Driven Challengers**

Emerging Product Capability, Best-in-class Business Capability

Companies with strong business presence and commercial capabilities, supported by established customer relationships, market reach, channels, or brand strength. However, their product portfolio may have relatively lower differentiation, breadth, maturity, or innovation compared with leading participants.

Trailblazers **Innovation-Driven Trailblazers**

Best-in-class Product Capability, Emerging Business Capability

Companies demonstrating strong product capabilities, innovation, or differentiated offerings, but with comparatively lower business scale or market penetration. Their growth potential depends on strengthening commercial execution, customer reach, partnerships, geographic presence, and overall market visibility.

Emerging Players **Foundation-Building Evolving**

Emerging Product Capability, Emerging Business Capability

Companies with developing product capabilities and relatively limited business presence. These participants may be at an earlier stage of market development, serve focused segments or geographies, and have opportunities to strengthen both their product proposition and commercial footprint.

Top 20 Companies

(plotted on the Quadrant Positioning chart)

BRAND	COMPANY	HQ	ROLE	QUADRANT	X (PRODUCT CAPABILITY)	Y (BUSINESS CAPABILITY)	OVERALL SCORE
Aspen Aerogels	Aspen Aerogels, Inc.	Northborough, Massachusetts, USA	Manufacturer	Leaders	95	98	96
AVD Fire	AVD Fire (subsidiary of Goodwin PLC)	London, United Kingdom	Manufacturer	Leaders	96	90	93
KULR	KULR Technology Group, Inc.	San Diego, California, USA	Manufacturer	Leaders	94	89	92
FCL-X	FCL-X Fire & Safety Inc.	Toronto, Canada	Manufacturer	Leaders	92	92	92
Pyrophobic Systems	Pyrophobic Systems Limited	Barrie, Canada	Manufacturer	Leaders	93	85	89
Fire Isolator	Fire Isolator BV	Oss, Netherlands	Manufacturer	Challengers	90	87	88
RETRON	RETRON GmbH	Lünen, Germany	Manufacturer	Challengers	88	83	86
Fire Cloak	Fire Cloak Limited	Sheffield, England	Manufacturer	Challengers	89	82	86
Brimstone	Brimstone Fire Protection	Plano, Texas, USA	Manufacturer	Challengers	87	83	85
Guard-K	Guard-K Co., Ltd.	Seoul, South Korea	Manufacturer	Challengers	90	78	84
Fike Blue	Fike Corporation	Blue Springs, Missouri, USA	Manufacturer	Trailblazers	98	81	90
Bridgehill	Bridgehill AS	Larvik, Norway	Manufacturer	Trailblazers	92	84	88
CellBlockEX	CellBlock FCS LLC	Standish, Maine, USA	Manufacturer	Trailblazers	93	82	88
AVL Stingray	AVL List GmbH	Graz, Austria	Manufacturer	Trailblazers	94	83	88
VLITEX	VLITEX GmbH	Bayreuth, Germany	Manufacturer	Trailblazers	93	79	86
LithiumSafe	LithiumSafe B.V.	Oosterhout, Netherlands	Manufacturer	Evolving Players	87	76	82
TL-X	TL-X	Goyang-si, South Korea	Manufacturer	Evolving Players	88	70	79
EV Fire Solutions	EV Fire Solutions Pty Ltd	NSW, Australia	Manufacturer	Evolving Players	80	76	78
LioShield	LioShield Safety Systems - FZCO	Dubai, UAE	Manufacturer	Evolving Players	82	66	74
Lion Battery	Lion Battery	Ahmedabad, India	Manufacturer	Evolving Players	67	75	71

Other Noticeable Players

(95 companies)

BRAND	COMPANY	HQ	ROLE	RELATIVE POSITIONING	STRENGTH
Ansul	Johnson Controls International plc	Cork, Ireland	Manufacturer	Leaders	
ZARGES	ZARGES GmbH	Weilheim, Germany	Manufacturer	Leaders	
DENIOS	DENIOS SE	Bad Oeynhausen, Germany	Manufacturer	Leaders	
Fire Rover	Fire Rover LLC	Farmington Hills, Michigan, USA	Manufacturer	Leaders	
HI-FOG	Marioff Corporation Oy (acquired by Inflexion)	Vantaa, Finland	Manufacturer	Leaders	
JUAND	Guo Anda Co., Ltd.	Xiamen, China	Manufacturer	Leaders	
AF-X Fireblocker	AF-X Systems B.V.	Purmerend, Netherlands	Manufacturer	Leaders	
AlkeGel	Alkegen	Irving, Texas, USA	Manufacturer	Leaders	
FirePro	FirePro Systems Ltd (subsidiary of Halma plc)	Limassol, Cyprus	Manufacturer	Leaders	
Stat-X	Fireaway Inc.	Minnetonka, Minnesota, USA	Manufacturer	Leaders	
Fogmaker	Fogmaker International AB (subsidiary of Dacke Industri)	Växjö, Sweden	Manufacturer	Leaders	
Aware Fire	Jiangxi Aware Fire Technology Co., Ltd.	Xinyu City, China	Manufacturer	Leaders	
Securiton	Securiton AG	Zollikofen, Switzerland	Manufacturer	Leaders	
SYSBEL	Shanghai SYSBEL Industry & Technology Co., Ltd.	Shanghai, China	Manufacturer	Leaders	
F-500 Encapsulator Agent	Hazard Control Technologies, Inc.	Fayetteville, Georgia, USA	Manufacturer	Leaders	
Minimax	Minimax Viking GmbH (subsidiary of Kirkbi A/S and Intermediate Capital Group plc)	Bad Oldesloe, Germany	Manufacturer	Leaders	
BEST	Rosenbauer International AG	Leonding, Austria	Manufacturer	Leaders	
WAGNER	WAGNER Group GmbH	Langenhagen, Germany	Manufacturer	Leaders	
Protecfire	Protecfire GmbH	Lübeck, Germany	Manufacturer	Leaders	
Dafo Vehicle	Dafo Vehicle AB (acquired by United Safety & Survivability Corporation)	Tyresö, Sweden	Manufacturer	Leaders	
Firetrace	Firetrace International (subsidiary of Halma plc)	Scottsdale, Arizona, USA	Manufacturer	Leaders	
LifeSafe	LifeSafe Technologies	Essex, UK	Manufacturer	Leaders	
ION-LINE	asecos GmbH	Gründau, Germany	Manufacturer	Leaders	
Ceasefire	Ceasefire Industries Pvt. Ltd.	Noida, India	Manufacturer	Leaders	
Dafo Vehicle Fire Suppression	Dafo Vehicle Fire Suppression AB (acquired by United Safety & Survivability Corporation)	Tyresö, Sweden	Manufacturer	Leaders	
Storemasta	Storemasta	Burnie, Tasmania, Australia	Manufacturer	Leaders	
Suntex	Suntex Composite Industrial Co., Ltd.	Zhenjiang, China	Manufacturer	Leaders	
Fireblock Lithium	Fireblock International (Pty) Ltd	Johannesburg, South Africa	Manufacturer	Leaders	
Victory Fire	Victory Fire & Gas Inc.	Houston, Texas, USA	Manufacturer	Challengers	
FlameBlock	FlameBlock (Pty) Ltd	Alberton, South Africa	Manufacturer	Trailblazers	
ANT	FIREKIM ENERGY SOLUTION Co., Ltd.	Cheongju, South Korea	Manufacturer	Leaders	
FIREKIM ES	FIREKIMES Co., Ltd.	Cheongju-si, South Korea	Manufacturer	Leaders	
Pyrogen	Pyrogen Fire Suppression Pty Ltd	Brisbane, Australia	Manufacturer	Challengers	

AFEX	AFEX Fire Suppression Systems	Raleigh, North Carolina, USA	Manufacturer	Leaders	
DSPA	DSPA B.V.	Beuningen, Netherlands	Manufacturer	Leaders	
Firechief	Firechief Global	Rutland, UK	Manufacturer	Leaders	
GreenSol	GreenEx Fire Suppression Solutions B.V.	Beek, Netherlands	Manufacturer	Leaders	
Padtex	Padtex Insulation	Valmiera, Latvia	Manufacturer	Leaders	
Rotarex Firetec	Rotarex S.A.	Lintgen, Luxembourg	Manufacturer	Leaders	
Brigit Fire	Brigit Fire Pty Ltd (subsidiary of Hudaco Industries)	Johannesburg, South Africa	Manufacturer	Challengers	
DIS Fire	DIS Fire Australia Pty Ltd	Sydney, New South Wales, Australia	Manufacturer	Challengers	
Kanex	Kanex Fire Solutions Limited	Mumbai, Maharashtra, India	Manufacturer	Leaders	
RSL Fire	RSL Fire B.V.	Almere, Netherlands	Manufacturer	Leaders	
Salgrom	Salgrom Technologies Oy	Tampere, Finland	Manufacturer	Leaders	
Texfire	Texfire Textils Tecnics S.L.	Barberà del Vallès, Spain	Manufacturer	Trailblazers	
Safex	Safex Fire Services Pvt. Ltd.	Mumbai, India	Manufacturer	Leaders	
Cold Fire	Firefreeze Worldwide Inc.	Rockaway, New Jersey, USA	Manufacturer	Trailblazers	
Firexo	Firexo Limited	London, UK	Manufacturer	Leaders	
BlazeCut	BlazeCut s.r.o.	Sydney, Australia	Manufacturer	Challengers	
MAUS	MAUS Industry AB	Stockholm, Sweden	Manufacturer	Leaders	
BAVARIA	Bavaria Egypt S.A.E.	Cairo, Egypt	Manufacturer	Evolving Players	
Firebox	Firebox Australia Pty Ltd	Beresfield, New South Wales, Australia	Manufacturer	Trailblazers	
Muster	MusterFire International Pty Ltd	Brisbane, Australia	Manufacturer	Trailblazers	
Anaf	Anaf Group S.p.A.	Gropparello, Italy	Manufacturer	Evolving Players	
Lifeguard	United Fire Equipments Pvt. Ltd.	New Delhi, India	Manufacturer	Trailblazers	
Britannia Fire	Britannia Fire Ltd.	Norwich, UK	Manufacturer	Trailblazers	
FireCape	Newtex Industries, Inc.	Victor, New York, USA	Manufacturer	Trailblazers	
Amerex	Amerex Corporation (subsidiary of McWane, Inc.)	Trussville, Alabama, USA	Manufacturer	Challengers	
Fire Engineering Technology	Fire Engineering Technology	New Delhi, India	Manufacturer	Trailblazers	
Hafex	Hafex Solutions	Heidelberg, South Africa	Manufacturer	Trailblazers	
Elect Tiger	Hancom Lifecare Inc.	Yongin, South Korea	Manufacturer	Challengers	
SafeQuip	Safequip (Pty) Ltd	Johannesburg, South Africa	Manufacturer	Evolving Players	
Ecosafe	ECOSAFE SA	Ecublens, Switzerland	Manufacturer	Trailblazers	
Promat	Etex Building Performance NV	Tisselt, Belgium	Manufacturer	Trailblazers	
PRI-SAFETY	Hangzhou PRI-SAFETY Fire Technology Co., Ltd.	Hangzhou, China	Manufacturer	Trailblazers	
LABNORI	LABNORI	Shenzhen, China	Manufacturer	Trailblazers	
COMBAT	Lingjack Engineering Works Pte Ltd	Singapore	Manufacturer	Evolving Players	
Supremex	Supremex Equipments Limited	Mumbai, India	Manufacturer	Trailblazers	
Relifire	Relifire India	Mumbai, India	Manufacturer	Trailblazers	
TLX Tech	TLX Technologies	Pewaukee, Wisconsin, USA	Manufacturer	Trailblazers	

Checkpoint Group	Checkpoint Group Pty Ltd	Sydney, New South Wales, Australia	Manufacturer	Evolving Players	
Modi International	Modi International	Delhi, India	Manufacturer	Evolving Players	
Reacton	Reacton Fire Suppression Ltd.	Chelmsford, UK	Manufacturer	Challengers	
NAFFCO	National Fire Fighting Manufacturing FZCO	Dubai, UAE	Manufacturer	Evolving Players	
Fire Fighter	Fire Fighter Industry Sdn Bhd	Petaling Jaya, Malaysia	Manufacturer	Evolving Players	
SFFECO	Saudi Factory for Fire Equipment Co.	Riyadh, Saudi Arabia	Manufacturer	Trailblazers	
Signature	Darshan Safety Zone	Ahmedabad, India	Manufacturer	Evolving Players	
Hazpak Fire	Hazpak (Pty) Ltd	Cape Town, South Africa	Manufacturer	Evolving Players	
HiCare	Hicare Protectivewear Pvt Ltd	Navi Mumbai, India	Manufacturer	Evolving Players	
Extinpel	Extinpel Extintores Ltda.	São Paulo, Brazil	Manufacturer	Evolving Players	
Gartec Safety	Gartec Safety Technology (Shenzhen) Co., Ltd.	Shenzhen, China	Manufacturer	Evolving Players	
Mocelin	Mocelin Indústria de Extintores Ltda.	Balsa Nova, Brazil	Manufacturer	Evolving Players	
New Pig Obsidian	New Pig Corporation	Tipton, Pennsylvania, USA	Manufacturer	Evolving Players	
Bristol	Bristol Fire Engineering	Dubai, UAE	Manufacturer	Evolving Players	
Hercules	Fire Armour Pte Ltd	Singapore	Manufacturer	Evolving Players	
Fire Terminator	Fire Terminator International Pte. Ltd.	Singapore	Manufacturer	Evolving Players	
JUTEC	JUTEC Biegesysteme GmbH	Limburg an der Lahn, Germany	Manufacturer	Trailblazers	
Morimi Fire	Morimi Fire (Pty) Ltd	Johannesburg, South Africa	Manufacturer	Evolving Players	
SRI	Steel Recon Industries Sdn Bhd	Subang Jaya, Malaysia	Manufacturer	Evolving Players	
Gilfire	Gilfire Equipamentos Contra Incêndio Ltda.	São Paulo, Brazil	Manufacturer	Evolving Players	
Safe Pro Fire	Safe Pro Fire Services Pvt. Ltd.	Mumbai, India	Manufacturer	Evolving Players	
Falcon Fire	Falcon Fire Pte. Ltd.	Singapore	Manufacturer	Evolving Players	
Fire Destroyer	Fire Destroyer	Riyadh, Saudi Arabia	Manufacturer	Evolving Players	
HeatSafe Systems	HeatSafe Systems	Lapstone, New South Wales, Australia	Manufacturer	Evolving Players	
Eversafe	Eversafe Extinguisher Sdn. Bhd.	Subang Jaya, Malaysia	Manufacturer	Evolving Players	

Parameter Detail by Company

(Product Capability and Business Capability evidence for the 20 companies on the chart)

Aspen Aerogels, Inc.

Aspen Aerogels · Leaders · X 95 · Y 98 · Overall 96

X — PRODUCT CAPABILITY

Fire Suppression Technology & Performance — scored 93/100

> The rating rests on the company's established, self-reported commercial deployments and regulatory compliance listings for its passive cell-to-cell thermal barriers. Based on primary research, the company has not disclosed specific third-party test results, such as explicit UL 9540A metrics, fire suppression times, or precise temperature reduction numbers.

SUPPORTING POINTS

- The company reports that its PyroThin cell-to-cell aerogel barriers function as passive fire barriers and compression pads to mitigate thermal runaway propagation by resisting heat transfer.
- The company reports that its PyroThin aerogel technology satisfies and exceeds China GB38031 and UN ECE/TRANS/180/Add.20 safety requirements for electric vehicle thermal runaway standards.

Battery Chemistry Compatibility & Coverage — scored 100/100

> The rating reflects the company's clear capability to tune its aerogel formulations for dominant lithium-ion chemistries and both pouch and prismatic cell form factors, verified through integration in major automotive lines like General Motors' Ultium platform. Based on primary research, the company has not disclosed specific compatibility lists or independent comparative test results for cylindrical form factors.

SUPPORTING POINTS

- The company reports that its PyroThin cell-to-cell thermal barriers are designed as a tunable platform to manage thermal runaway propagation across LFP, mid-nickel, NMC, and solid-state battery chemistries, and the company operates a dedicated Global Advanced Thermal Barrier Center near Boston collaborating on these cell chemistries.
- The company reports that its PyroThin cell-to-cell thermal barriers are engineered specifically for pouch and prismatic cell formats to act simultaneously as a fire barrier and mechanical compression pad.

Product Portfolio Breadth & Modularity — scored 89/100

> We verified that the company demonstrates clear capability in passive thermal insulation barriers through its configurable PyroThin platform, which scales from individual cell configurations to module and pack barriers across automotive and energy storage formats. Based on primary research, the company has not disclosed active fire safety products such as cell-level sensors or active pack-level chemical suppression systems within its battery safety portfolio.

SUPPORTING POINTS

- The company reports that its PyroThin passive thermal barrier portfolio covers a range of thickness and stiffness profiles engineered to prevent cell-to-cell thermal runaway propagation across multiple cell formats, chemistries, and module/pack-level architectures.
- Product documentation specifies distinct performance grades, including the ATB1000 for maximum thermal resistance and ATB2000 for enhanced compression performance, available in standard modular thicknesses of 2mm and 3mm.

R&D and Innovation Pipeline — scored 100/100

> The rating rests on the company's dedicated rapid prototyping infrastructure at its Advanced Thermal Barrier Center and active battery-focused patent activity for its commercialized PyroThin technology. Based on primary research, the company has not disclosed its exact annual research and development spend allocated specifically to the battery fire safety segment versus its energy industrial product line.

SUPPORTING POINTS

- The company opened its 59,000-square-foot Advanced Thermal Barrier Center (ATBC) in Marlborough, Massachusetts in June 2023, serving as an engineering and rapid prototyping hub to develop next-generation PyroThin cell-to-cell thermal mechanical barrier technologies for electric vehicles and energy storage system applications.
- The company reports active technology scaling within its battery materials segment, securing 34 battery-related patents during the second quarter of 2024 for innovations including fiber-reinforced aerogel material with improved durability and thermal performance.

Certifications & Standards Compliance — scored 92/100

> The assessment rests on the company's verified automotive quality certifications and its reported compliance with regional EV safety mandates such as China's GB 38031 standards. Based on primary research, the company has not disclosed formal third-party certificates or test reports showing compliance with standard BESS fire safety codes like UL 9540A, NFPA 855, or IEC 62619, although it actively recruits personnel to manage these specific qualifications.

SUPPORTING POINTS

- Aspen Aerogels reports that its PyroThin aerogel cell-to-cell thermal barriers satisfy or exceed China GB 38031 and UN ECE/TRANS/180/Add.20 safety requirements.
- The company demonstrates compliance with automotive-specific quality standards, having officially achieved IATF 16949 certification for one of its manufacturing facilities, alongside an ISO 9001 certification.

Y — BUSINESS CAPABILITY

Named Customers & Contracts — scored 100/100

> We verified that the company has a strong baseline of publicly declared multi-year contracts and Tier 1 design awards with prominent global automotive manufacturers for its cell-to-cell thermal barriers. Based on primary research, the company has not disclosed the precise contract values, delivery volumes, or pricing terms for most of these original equipment manufacturer partnerships.

SUPPORTING POINTS

- Aspen Aerogels, Inc. has entered into multi-year production contracts with General Motors to supply multi-part PyroThin thermal barriers for its next-generation electric vehicles through terms expiring from 2026 to 2034.
- The company has secured supply contracts and design awards with Toyota, Scania, and Automotive Cells Company (ACC), as well as a contract via manufacturing partner Valmet Automotive to supply thermal barriers for the electric Porsche 718 series.

Revenue & Growth — scored 100/100

> We verified that the rating is anchored by the company's segment financial reporting detailing explosive annual growth for its battery thermal barrier systems through 2024, followed by shorter-term fluctuations tied to broader EV market demand shifts. Based on primary research, the company has not disclosed the precise year-over-year growth percentage or finalized full-year breakdown for the specific product segment in 2025.

SUPPORTING POINTS

- The company reports that its Thermal Barrier segment, which includes the PyroThin aerogel thermal barriers developed for mitigation of thermal runaway in EV battery packs, generated annual product revenues of $306.8 million in 2024, up from $110.1 million in 2023 and $55.6 million in 2022. TradingView
- The company reports that its total quarterly Thermal Barrier segment revenue was $29.5 million for the second quarter of 2026, marking an 81% increase quarter-over-quarter compared to the first quarter of 2026.

Geographic Footprint & Channel Network — scored 92/100

> The rating rests on the company's verified deployment of thermal runaway solutions to major regional automotive platforms across North America, Europe, and Asia, backed by domestic and international production sites. Based on primary research, the company has not disclosed the exact number of countries served or specific distribution sales partners for its battery fire safety segment.

SUPPORTING POINTS

- The company reports a global manufacturing and R&D footprint for its thermal barrier business consisting of corporate offices and research labs in Massachusetts, a manufacturing plant in Rhode Island, and a high-volume automated PyroThin product assembly facility in Apodaca, Mexico.
- The company reports that its PyroThin thermal runaway barriers are in volume production for major automotive OEMs across three primary geographic regions: North America, Europe, and Asia.

Installed Base & Market Share — scored 100/100

> We verified that the company has achieved extensive adoption of its aerogel-based barriers across major automotive platforms, indicating a large active footprint in the electric vehicle fire containment market. Based on primary research, the company has not disclosed its precise global market share percentage or exact installed unit counts deployed in the field.

SUPPORTING POINTS

- The company reports more than USD 300 million in electric vehicle thermal barrier revenue during 2024, driven primarily by its PyroThin cell-to-cell barriers.
- The company reports volume production contracts with major automotive manufacturers, including General Motors, Toyota, Audi, Scania, and Automotive Cells Company.

Funding & Strategic Partnerships — scored 100/100

> We verified that the company possesses extensive, market-leading capital backing and strategic commercial alliances with major automotive OEMs and battery cell manufacturers for its anti-thermal runaway aerogel barriers. Based on primary research, the company has not disclosed the specific pricing frameworks or minimum volume obligations under its long-term automotive supply contracts.

SUPPORTING POINTS

- Aspen Aerogels, Inc. received a conditional commitment for a $670.6 million loan from the U.S. Department of Energy Loan Programs Office in October 2024 to finance the construction of its PyroThin aerogel blanket manufacturing plant in Register, Georgia.
- The company entered into multi-year commercial production contracts and a $100 million secured loan agreement with General Motors, alongside supply agreements with Toyota, Scania, and Automotive Cells Company to provide PyroThin thermal barriers for electric vehicle battery systems.

AVD Fire (subsidiary of Goodwin PLC) AVD Fire ● Leaders X 96 Y 90 Overall 93

X — PRODUCT CAPABILITY

Fire Suppression Technology & Performance — scored 98/100

> We verified that the company's proprietary technology has third-party certification under European and Dutch battery fire test protocols for portable systems. Based on primary research, the company has not disclosed large-scale UL 9540A rack- or unit-level suppression time and exact temperature reduction metrics for industrial battery energy storage systems.

SUPPORTING POINTS

- The company reports its AVD (Aqueous Vermiculite Dispersion) fire extinguishing agent has successfully type-tested on a 6.8 kWh battery module to extinguish flames and prevent thermal propagation.
- AVD Fire has achieved independent compliance and testing verification under the Kiwa Test Protocol NTA 8133 for portable fire extinguishers on lithium-ion battery fires, alongside ZSW testing in Germany.

Battery Chemistry Compatibility & Coverage — scored 98/100

> The rating rests on the company's clear mapping of chemistry compatibility across five major lithium-ion formulations paired with supporting independent academic test data evaluating diverse cell form factors like pouch and cylindrical cells. Based on primary research, the company has not disclosed comprehensive, standardized application notes specifying performance constraints by exact watt-hour limits across all forms.

SUPPORTING POINTS

- The company reports that its Aqueous Vermiculite Dispersion (AVD) technology has undergone performance testing and is validated for lithium battery types including lithium iron phosphate (LFP), nickel manganese cobalt oxide (NMC), lithium cobalt oxide (LCO), lithium manganese oxide (LMO), and nickel cobalt aluminium oxide (NCA).
- Independent third-party experimental testing published in 2020 evaluated the efficiency of the AVD fire extinguishing agent on NMC pouch cells, LFP pouch cells, and 21700 cylindrical cells, verifying a maximum cooling rate of 84.450 °C/s for NMC pouch cells and 202.440 °C/s for LFP pouch cells.

Product Portfolio Breadth & Modularity — scored 94/100

> We verified that the firm maintains a clear and diverse portfolio of certified, deployed handheld and mobile suppression hardware utilizing its proprietary chemical agent across public, transport, and commercial settings. Based on primary research, the company has not disclosed cell-level sensor integrations or granular modular pack-level management mechanisms within its standard technical documentation.

SUPPORTING POINTS

- AVD Fire reports a commercial product portfolio centered around its Aqueous Vermiculite Dispersion (AVD) technology, which includes handheld extinguishers (ranging from 500 ml aerosol cans to 2L, 6L, and 9L capacities), 25L and 50L mobile trolley-mounted canisters, specialized fire blankets, and fire-resistant containers.
- The company reports its technology is scalable for use in fixed fire safety system installations (such as sprinkler setups) alongside its portable delivery systems.

R&D and Innovation Pipeline — scored 96/100

> The rating reflects the company's clear capability in introducing certified product lines like EV fire blankets and specialty portable suppression hardware leveraging its proprietary mineral agent formulation. Based on primary research, the company has not disclosed its exact R&D spending figures, patent filing numbers, or active projects mapping out integration with electronic battery management systems.

SUPPORTING POINTS

- AVD Fire developed Aqueous Vermiculite Dispersion (AVD), a proprietary fire extinguishing agent composed of chemically exfoliated vermiculite platelets suspended in water specifically engineered to cool cells and form an oxygen barrier to mitigate lithium-ion thermal runaway.
- AVD Fire launched a new DIN-specification certified electric vehicle (EV) fire blanket and showcased its NTA 8133-certified 6-litre and 9-litre portable fire extinguishers at the Intersec Dubai expo in January 2026.

Certifications & Standards Compliance — scored 95/100

> The rating rests on the company's verified third-party compliance with portable extinguisher and vehicle fire blanket standards like NTA 8133 and DIN Spec 91489:2024-11. Based on primary research, the company has not disclosed compliance or large-scale testing data under utility-scale ESS or stationary battery standards such as UL 9540A or NFPA 855.

SUPPORTING POINTS

- The company reports its 6-litre and 9-litre portable fire extinguishers are certified to NTA 8133, which is the internationally recognized Kiwa test standard for lithium-ion battery fire performance.
- The company reports its EV Fire Blanket is independently tested and certified to the DIN Spec 91489:2024-11 standard by KIWA MPA Dresden.

Y — BUSINESS CAPABILITY

Named Customers & Contracts — scored 89/100

> The assessment rests on the company's regional distribution and public network installations. Based on primary research, the company has not disclosed specific commercial contract values or named battery manufacturing or electric vehicle OEM customers.

SUPPORTING POINTS

- The company reports its technology has achieved Qatari Civil Defence approval and Dubai Metro network blanket installations.
- AVD Fire announced an official UK distribution partnership with Sentura Group / FireChief in March 2022.

Revenue & Growth — scored 95/100

> The assessment rests on partial financial disclosures released by the parent entity, Goodwin PLC, which confirms a 30% year-on-year profit expansion for the business unit alongside Dupré Minerals. Based on primary research, the company has not disclosed standalone, isolated annual revenue figures or granular year-over-year revenue growth percentages specifically for the AVD Fire brand.

SUPPORTING POINTS

- The parent company, Goodwin PLC, reported that combined profits from AVD Fire and Dupré Minerals increased by 30% year-on-year to £1.0 million for the period leading up to September 2026.
- The company's parent reported overall continuing business revenue of £68.9 million and an operating profit of £10 million across its combined Refractory and Technological divisions (which include AVD Fire) for the financial year ended 30 April 2026.

Geographic Footprint & Channel Network — scored 92/100

> The assessment rests on the company's verified primary office in the United Kingdom and its extensive, rapidly expanding network of authorized international distribution partners spanning Europe, Asia, the Middle East, and South America to deliver its specialized battery fire suppression products. Based on primary research, the company has not disclosed the exact total number of active international office locations or direct employee counts dedicated to regional support channels.

SUPPORTING POINTS

- The company reports adding 10 new distributors across 10 countries (including Lithuania, India, Pakistan, Estonia, Latvia, Ukraine, Bulgaria, Serbia, Peru, and Trinidad & Tobago) to its global network, augmenting established regional partnerships in Spain, Taiwan, Australia, Thailand, and the United Arab Emirates.
- The company maintains its registered corporate office in Stoke-on-Trent, United Kingdom, and is established as a subsidiary operating unit of Goodwin PLC.

Installed Base & Market Share — scored 94/100

> The assessment rests on the company's confirmed public deployment milestones, such as its transit infrastructure installations in Dubai. Based on primary research, the company has not disclosed its precise global installed unit counts or its exact percentage of market share within the battery fire safety industry.

SUPPORTING POINTS

- The company reports blanket installations of its fire protection systems across the Dubai Metro network.

Funding & Strategic Partnerships — scored 82/100

> We verified that the company's market presence relies on commercial distribution agreements and internal corporate backing from its parent company, Goodwin PLC, rather than independent venture capital or institutional funding. Based on primary research, the company has not disclosed any strategic alliances or joint ventures with battery manufacturers or research institutions.

SUPPORTING POINTS

- AVD Fire announced an official UK distribution partnership with Sentura Group / FireChief in March 2022.
- Classic Fire + Life Safety announced a partnership with Victory Fire and Gas in August 2024 to distribute AVD fire extinguishers in Canada.

KULR Technology Group, Inc. KULR ● Leaders X 94 Y 89 Overall 92

X — PRODUCT CAPABILITY

Fire Suppression Technology & Performance — scored 98/100

> We verified that the company possesses extensive, third-party aligned validation for its passive propagation prevention and thermal management architectures through international standards testing and federal research partnerships. Based on primary research, the company has not disclosed specific quantitative metrics detailing fire suppression turnaround times or direct chemical agent suppression speeds under this individual parameter.

SUPPORTING POINTS

- The company reports that its passive propagation resistant battery solution was relied upon by an industrial-scale energy storage customer in successfully meeting and passing the safety thresholds established for UL 9540 and UL 9540A certification.
- The company reports that its Thermal Runaway Shield technology prevents cell-to-cell thermal runaway propagation by passively absorbing heat via a carbon fiber core and integrated liquid coolant sleeve, containing the failure area and safely venting gases without releasing flames.

Battery Chemistry Compatibility & Coverage — scored 93/100

> The assessment rests on the company's established capability to engineer passive propagation resistant enclosures and testing methodologies for cylindrical form factors and its extension into prismatic designs. Based on primary research, the company has not disclosed a standardized, cross-referenced compatibility matrix or comprehensive test datasets confirming product performance specifically across varied lithium-ion chemistries like LFP versus NMC.

SUPPORTING POINTS

- The company reports that its Thermal Runaway Shield (TRS) technology, originally designed for cylindrical cells using carbon fiber wicks and vaporizing heat sinks, has been expanded via a patent application to be molded to fit various prismatic cell geometries.
- The company reports that its proprietary KULR ONE Design Solutions (K1DS) platform provides cell-level characterization and testing protocols—including Fractional Thermal Runaway Calorimetry—to analyze thermal runaway behaviors across customer-specified battery cells.

Product Portfolio Breadth & Modularity — scored 100/100

> The rating rests on the company's clear and diverse commercial range of modular battery architectures and passive propagation resistant products spanning from cell-level sleeves to pack-level containment boxes and software systems. Based on primary research, the company has not disclosed standard product catalog datasheets detailing exact modular dimensions and scaling limitations for individual customized commercial integrations.

SUPPORTING POINTS

- The company reports its KULR ONE platform as a modular, integrated power stack spanning cell/pack design, battery management systems (kBMS), and thermal runaway protections. Its modular battery portfolio encompasses the K1S Space battery series (ranging from 100 Wh to 400 Wh), K1M MAX data center backups, and SafeCASE shipping containers.
- The company reports cell-level fire safety offerings including the HYDRA Thermal Runaway Shield (TRS)—a patented sleeve-like shield with a carbon fiber core and liquid coolant that surrounds individual cells—alongside cell screening and licensed NASA Internal Short Circuit (ISC) trigger devices for controlled laboratory cell failure testing.

R&D and Innovation Pipeline — scored 86/100

> The assessment rests on the company's verified portfolio of battery safety patents and its launch of next-generation product architectures integrating thermal protection with battery management systems. Based on primary research, the company has not disclosed its exact dollar investments in research and development for new fire safety technologies.

SUPPORTING POINTS

- KULR Technology Group, Inc. reports the expansion of its intellectual property portfolio through the award of multiple U.S. patents, including U.S. Patent No. 11,018,397 covering its Thermal Runaway Shield (TRS) technology designed to passively reduce fire and explosion risks in lithium-ion battery packs.
- The company reports new product innovation and expansion through the launch of its next-generation, analog-architecture Battery Management System (kBMS) engineered for safety and radiation tolerance, alongside the commercialization of its KULR ONE modular energy platform featuring integrated thermal runaway suppression materials.

Certifications & Standards Compliance — scored 95/100

> The rating rests on the company's demonstrated capability to enable its commercial clients to successfully pass UL 9540 and UL 9540A safety test thresholds using its integrated thermal containment solutions. Based on primary research, the company has not disclosed specific standalone direct product listings or certifications under NFPA 855 or IEC 62619 for its own off-the-shelf component inventory.

SUPPORTING POINTS

- The company reports that its Passive Propagation Resistant (PPR) solution was utilized by an industrial-scale energy storage system manufacturing customer to design, test, and achieve the stringent UL 9540 (Field-Listed) and UL 9540A (Cell/Module/Pack-Tested) safety thresholds.
- KULR Technology Group, Inc. reports that it maintains an AS9100 certified quality management system and performs in-house battery stress testing against MIL-SPEC and UL standards.

Y — BUSINESS CAPABILITY

Named Customers & Contracts — scored 98/100

> We verified that the company has a strong record of securing defense, aerospace, and commercial energy contracts, including multi-million dollar deals and long-term supply agreements. Based on primary research, the company has not disclosed the specific names of multiple commercial drone and industrial-scale energy storage customers referenced in its recent contract wins.

SUPPORTING POINTS

- The company reports a preferred battery supply agreement announced in 2026 with Caban Energy estimated to generate $30 million in total revenue over five years.
- The company reports a follow-on contract announced in 2023 valued at $2.68 million as the exclusive supplier of fail-safe battery safety solutions for an industrial scale energy storage system manufacturer.

Revenue & Growth — scored 95/100

> We verified the company's reported full-year 2025 revenues and its 51% annual growth rate through its financial statements. Based on primary research, the company has not disclosed the exact revenue breakdown or specific growth percentages derived exclusively from battery fire safety products and services versus its overall thermal management and energy storage operations.

SUPPORTING POINTS

- The company reports record annual revenues of $16.2 million for 2025, which represents a 51% increase compared to total revenues of $10.7 million for 2024.
- The company reports a 39% year-over-year growth in product revenue for the full year 2025 as part of its strategic shift toward a product-led commercial model.

Geographic Footprint & Channel Network — scored 84/100

> The rating rests on the company's operational footprint being concentrated entirely within the United States to serve domestic aerospace, defense, and industrial markets. Based on primary research, the company has not disclosed the number of countries served, external distribution networks, or third-party global sales partners.

SUPPORTING POINTS

- The company reports that its corporate headquarters, engineering center ("Forge-1"), and volume manufacturing facility ("Forge-2") are located in Webster, Texas, alongside additional engineering operations in San Diego, California.
- The company reports a specialized supply chain focusing on end-to-end domestic manufacturing and cell sourcing from allied nations under National Defense Authorization Act (NDAA) guidelines.

Installed Base & Market Share — scored 76/100

> We verified that the company relies primarily on individual commercial and defense program contract updates and technology licensing agreements to document its deployment footprint. Based on primary research, the company has not disclosed its exact total installed system counts or its defined percentage share of the global battery fire safety market.

SUPPORTING POINTS

- The company reports that it has secured a partnership with Americase, which holds a 70% market share in damaged, defective, or recalled battery shipments, exclusively licensing KULR Technology Group, Inc.’s passive propagation resistant (PPR) technology.
- The company reports it has an automated battery cell testing platform designed with an initial processing capability of 500,000 cells annually.

Funding & Strategic Partnerships — scored 90/100

> We verified that the company has established multiple strategic development partnerships with battery manufacturers and integration partners to deploy its thermal management and safety engineering architectures across aerospace, defense, and commercial markets. Based on primary research, the company has not disclosed the specific monetary values or formal institutional investor backing details associated with external funding rounds for this market category.

SUPPORTING POINTS

- KULR Technology Group, Inc. reports a strategic partnership with Worksport Ltd. announced in February 2025 focusing on joint battery pack development, thermal runaway protection, and AI-integrated battery management software design.
- KULR Technology Group, Inc. reports a joint development collaboration announced in December 2025 with a global battery-cell manufacturer to co-develop a next-generation KULR ONE MAX Battery Backup Unit product line for data centers.

FCL-X Fire & Safety Inc. FCL-X ● Leaders X 92 Y 92 Overall 92

X — PRODUCT CAPABILITY

Fire Suppression Technology & Performance — scored 98/100

> The rating rests on the company's verified compliance with the Dutch NTA 8133 standard and its UL component recognition for Class A fires. Based on primary research, the company has not disclosed independent UL 9540A test results, exact suppression times, or specific temperature reduction metrics.

SUPPORTING POINTS

- The company reports that its water-based PFAS-free lithium-ion fire extinguishing agent has been certified to the Dutch NTA 8133 standard designation for lithium-ion battery fires.
- The company reports that its extinguishing agent has received Class A Fire Water-Based Agent Fire Extinguishers Component Recognition from Underwriters Laboratories Solutions (UL).

Battery Chemistry Compatibility & Coverage — scored 91/100

> The evaluation is based on independent certification to the Dutch NTA 8133 standard for lithium-ion battery fire mitigation and self-reported broad application categories spanning consumer electronics to electric vehicles. Based on primary research, the company has not disclosed comprehensive compatibility matrices, explicit test data delineating performance by battery chemistry, or form-factor application notes.

SUPPORTING POINTS

- The company reports that its FCL-X water-based, PFAS-free fire extinguishing agent is purpose-built to stop thermal runaway and extinguish fires in lithium-ion battery cells and pack structures across varying device categories like power tools, e-mobility devices, consumer electronics, and electric vehicles.
- We verified through testing records that the FCL-X agent achieved the Dutch NTA 8133 standard designation for lithium-ion battery fire extinguishers, though specific test metrics across distinct chemistries like NMC or LFP and form factors like cylindrical, prismatic, or pouch remain unstated.

Product Portfolio Breadth & Modularity — scored 89/100

> The rating rests on the company's documented commercial capability to deliver a single specialized fire suppression agent across multiple physical formats, ranging from portable consumer devices up to high-capacity wheeled industrial machinery and specialized gel variants. Based on primary research, the company has not disclosed cell-level hardware tracking or integrated pack-level hardware multi-sensor monitoring arrays.

SUPPORTING POINTS

- The company reports offering its proprietary FCL-X water-based fire extinguishing agent in multiple delivery formats, including handheld consumer sizes (20 oz., 1-litre, and 2-litre models), commercial sizes (2.5-gallon units), and wheeled industrial configurations (50-litre and 100-litre models).
- The company reports developing FCL-X GEL, a next-generation gel-based formulation with advanced surface adhesion designed specifically to scale for large-scale high-energy systems like Battery Energy Storage Systems (BESS) and industrial energy infrastructure.

R&D and Innovation Pipeline — scored 90/100

> We verified that the company has established verifiable capability through the active development of specialized lithium-ion suppression variations like its battery gel agent and has pursued regional patent protection. Based on primary research, the company has not disclosed its exact annual research and development expenditures or a structured multi-year innovation roadmap.

SUPPORTING POINTS

- The company reports the ongoing development of its next-generation FCL-X™GEL technology specifically formulated for battery energy storage systems (BESS).
- The company has submitted patent applications to protect its product innovations across North America, including Canada, the United States, and Mexico.

Certifications & Standards Compliance — scored 92/100

> The assessment rests on the company's confirmed UL Component Recognition for Class A fires and third-party validation under the Dutch NTA 8133 standard. Based on primary research, the company has not disclosed formal, direct product listings or certifications under UL 9540A, NFPA 855, or IEC 62619 standards for battery fire safety.

SUPPORTING POINTS

- The company reports its FCL-X fire extinguishing agent achieved UL Class A Fire Water-Based Agent Fire Extinguishers Component Recognition for the United States and Canada in July 2025.
- The company reports that its agent has been tested by SafeLabs and KIWA, is certified to the Dutch NTA 8133 standard for lithium-ion battery fire suppression, and has been evaluated through testing protocols mirroring UL 9540A and UL 711.

Y — BUSINESS CAPABILITY

Named Customers & Contracts — scored 93/100

> The evaluation is based on confirmed commercial activity with a prominent automotive manufacturer for facility safety services and reported fire-extinguishing agent sales to a global electric vehicle original equipment manufacturer. Based on primary research, the company has not disclosed the specific monetary values of these agreements or the identity of the global electric vehicle original equipment manufacturer.

SUPPORTING POINTS

- FCL-X Fire & Safety Inc. executed an Integrated Safety Purchase Order (ISPO) announced on September 29, 2026, with Kia Georgia, Inc. for its West Point, Georgia facility to provide construction support and Health, Safety, and Environmental (HSE)-related services, including risk assessment, fire protection engineering, and implementing safety protocols for battery-related environments.
- The company reported securing sales of its FCL-X lithium-ion battery fire-extinguishing agent to a global electric vehicle original equipment manufacturer (EV OEM) customer for its EV plant in Georgia, as announced on June 4, 2024.

Revenue & Growth — scored 100/100

> The rating rests on the verified financial disclosures of the public entity confirming expanding commercial revenues specifically tied to its specialized fire safety solutions and agents. Based on primary research, the company has not disclosed the precise granular breakdown of revenue streams split strictly between retail extinguisher sales, bulk industrial concentrate supply, and on-site support services.

SUPPORTING POINTS

- For the third quarter ended July 31, 2026, the company reported sales of USD 0.156539 million compared to USD 0.04303 million a year ago, and for the nine months ended July 31, 2026, sales were USD 0.310031 million compared to USD 0.633017 million in the prior year period.
- In the fiscal year ended October 31, 2025, the company's total annual revenue was USD 693,999, compared to USD 9,990 in fiscal year 2024.

Geographic Footprint & Channel Network — scored 87/100

> The evaluation is based on the company's verified commercial presence and distribution partnerships established across the United States and Canada. Based on primary research, the company has not disclosed the specific total number of global offices, international distributors, or sales partners outside of North America.

SUPPORTING POINTS

- The company reports that its products are sold primarily in the United States and Canada, with patent applications submitted for North America covering Canada, the United States, and Mexico.
- The company reports that its distribution network includes partner relationships with commercial suppliers such as US Fire Pump, Firetron Inc., and Howell Rescue Systems.

Installed Base & Market Share — scored 94/100

> We verified that the company has initiated commercial adoption among public emergency services and secured an infrastructure service contract with a major automotive manufacturer. Based on primary research, the company has not disclosed its specific installed unit counts or its exact percentage of market share within the battery fire safety industry.

SUPPORTING POINTS

- The company reports growing customer adoption and states that its specialty lithium-ion battery fire-extinguishing technology has been adopted by several U.S. fire departments and electric vehicle manufacturers.
- The company executed an Integrated Safety Purchase Order with Kia Georgia, Inc. in September 2026 for construction support and health, safety, and environmental-related services at its West Point, Georgia facility.

Funding & Strategic Partnerships — scored 88/100

> The company's assessment rests on its verified strategic commercial partnerships and services agreement with an automotive manufacturer for battery infrastructure safety, alongside a national retail distribution arrangement for its specialty extinguishing units. Based on primary research, the company has not disclosed external venture funding rounds, institutional investment totals, or joint venture capital equity alliances.

SUPPORTING POINTS

- FCL-X Fire & Safety Inc. executed an Integrated Safety Purchase Order with Kia Georgia, Inc. announced on September 29, 2026, to provide specialized safety infrastructure, construction support, risk assessment, and fire protection engineering services for battery environments at its West Point, Georgia facility.
- FCL-X Fire & Safety Inc. reports a strategic commercial partnership to distribute its purpose-built lithium-ion battery fire extinguishers on a major national home improvement retail platform starting in 2026.

Pyrophobic Systems Limited Pyrophobic Systems ● Leaders X 93 Y 85 Overall 89

X — PRODUCT CAPABILITY

Fire Suppression Technology & Performance — scored 93/100

> The evaluation is based on the company's verified testing alignment with aerospace and maritime validation bodies for its passive intumescent barriers designed to prevent cell-to-cell thermal propagation. Based on primary research, the company has not disclosed specific suppression times, quantitative temperature reduction metrics, or full testing data sheets from their UL 9540A evaluations.

SUPPORTING POINTS

- The company reports that its LithiumPrevent passive fire containment materials, which leverage a proprietary intumescent polymer, have been tested and evaluated by NASA, the FAA, and DNV, and are proven to satisfy fire test standards including UL 9540A.

Battery Chemistry Compatibility & Coverage — scored 93/100

> Our assessment rests on the company's self-reported versatility across all major cell form factors and its stated compatibility with both LFP and ternary chemistries. Based on primary research, the company has not disclosed comprehensive, third-party verified compatibility matrices or comparative test datasets detailing performance differences across varying advanced battery chemistries.

SUPPORTING POINTS

- The company reports that its LithiumPrevent technology is applicable across all primary lithium battery configurations, including cylindrical, pouch, and prismatic form factors.
- The company reports that its LithiumPrevent 200REX formulation is designed for application with lithium iron phosphate (LFP) and ternary lithium cells.

Product Portfolio Breadth & Modularity — scored 89/100

> The evaluation rests on the company's established, commercially integrated passive fire protection material portfolio that offers modular design flexibility through injection molding or extrusion for cell and module containment. Based on primary research, the company has not disclosed any cell-level active sensors or pack-level active chemical suppression systems within its product line.

SUPPORTING POINTS

- The company reports its LithiumPrevent product line consists of patented intumescent thermoplastics (including LP200, LP200R, and LP200REX formulations) that can be injection molded or extruded into custom 3D pack components such as cell holders, spacers, busbar caps, module separators, and pack lids to isolate and contain thermal runaway at the cell and module level.
- We verified through a commercial announcement that General Motors selected the LithiumPrevent 200 material to replace traditional glass-filled plastic components within its Ultium electric vehicle battery platform.

R&D and Innovation Pipeline — scored 96/100

> The rating rests on the company's confirmed commercialization of patented intumescent thermoplastic formulations designed specifically for lithium-ion cell isolation and its verified engineering collaboration with major automotive OEMs. Based on primary research, the company has not disclosed its exact R&D spending figures or active pipeline investments into advanced detection algorithms and direct battery management system integration.

SUPPORTING POINTS

- The company reports owning 9 patents and has commercialized its LithiumPrevent family of formulations, including LP200R injection molding, LP200REX extrusion, and LithiumPrevent-2025 platforms designed for cell and module-level lithium-ion battery fire containment.
- General Motors selected the company's LithiumPrevent 200 material for application in its Ultium battery platform following a two-year collaborative testing and certification process.

Certifications & Standards Compliance — scored 93/100

> The assessment rests on the company's internal reports and validation by entities like NASA and the FAA meeting UL 9540A test methods for its passive containment components. Based on primary research, the company has not disclosed independent third-party listing certificates or test reports from nationally recognized testing laboratories directly certifying their custom battery hardware under NFPA 855 or IEC 62619.

SUPPORTING POINTS

- The company reports that its LithiumPrevent proprietary intumescent thermoplastic polymer has been tested and evaluated by NASA, the FAA, and DNV, and meets fire testing standards including UL 9540A.
- Pyrophobic Systems Limited was awarded ISO 9001:2015 certification on April 10, 2024, for its quality management system covering the manufacture of intumescent polymer products and fire safety solutions for battery energy storage and electric vehicle applications.

Y — BUSINESS CAPABILITY

Named Customers & Contracts — scored 100/100

> The rating rests on a confirmed multi-year component supply selection by a major automotive OEM alongside an active engineering partnership. Based on primary research, the company has not disclosed the exact contract values or specific delivery volumes for these agreements.

SUPPORTING POINTS

- General Motors selected Pyrophobic Systems' LithiumPrevent 200 fire safety material for its Ultium battery platform under a program running until 2030, following a two-year engineering collaboration to test and certify the material for safety and compliance, announced in February 2023.
- Acculon Energy announced a partnership with Pyrophobic Systems in September 2023 to provide design, prototyping, certification, and manufacturing services for battery systems.

Revenue & Growth — scored 73/100

> Based on primary research, the company has not disclosed annual revenue and growth rate from battery fire safety products and services.

SUPPORTING POINTS

- Not available

Geographic Footprint & Channel Network — scored 76/100

> The evaluation is based on the company's single confirmed physical headquarters and manufacturing site in Canada, supplemented by strategic automotive tier partnerships in North America. Based on primary research, the company has not disclosed the exact number of international offices, proprietary distribution hubs, or active third-party sales partners outside its primary Canadian location.

SUPPORTING POINTS

- The company reports its corporate headquarters and in-house production facilities are based at 649 Welham Road in Barrie, Ontario, Canada.
- The company reports supply partnerships with tier 1 and tier 2 automotive industry suppliers, alongside a partnership with a "Big 3" North American automobile manufacturer to supply its LithiumPrevent material.

Installed Base & Market Share — scored 85/100

> We verified that the company has secured a commercial supply contract with a major automotive platform for its passive thermal containment material, indicating real-world field deployment. Based on primary research, the company has not disclosed its total installed unit counts, precise shipment numbers, or global battery safety market share.

SUPPORTING POINTS

- In February 2023, the company announced that General Motors selected its LithiumPrevent 200 material for deployment in the Ultium electric vehicle battery platform.
- Not available Strategic Market Research

Funding & Strategic Partnerships — scored 90/100

> We verified that the company possesses established commercial development partnerships and technical collaborations with major automotive and energy storage tier suppliers, though no external venture capital or private equity funding rounds have been confirmed in public disclosures. Based on primary research, the company has not disclosed any institutional investment history or raised funding rounds.

SUPPORTING POINTS

- Pyrophobic Systems Limited announced a strategic partnership with Acculon Energy in September 2023 to combine its proprietary intumescent polymers with Acculon's battery design, prototyping, and certification services to address fire safety and thermal runaway containment.
- The company reports a formal, multi-year collaboration established in February 2023 with General Motors, where they combined forces with Moulded Precision Components (MPC) to test and certify their LithiumPrevent 200 material for use in GM's Ultium battery platform through a program scheduled to run until 2030.

Fire Isolator BV Fire Isolator ● Challengers X 90 Y 87 Overall 88

X — PRODUCT CAPABILITY

Fire Suppression Technology & Performance — scored 90/100

> The assessment rests on the company's internal live test data and basic material compliance certifications for its multi-component electric vehicle fire containment system. Based on primary research, the company has not disclosed standardized third-party battery-level performance testing results, such as UL 9540A metrics, or independent verification of its suppression time.

SUPPORTING POINTS

- The company reports that covering an actively burning electric vehicle with its fire blanket decreases fire temperatures from 1,500 °C to approximately 600–800 °C, which drop further to 200–300 °C after applying its aerosol units and water mist lance.
- The company reports its fire blanket is certified under ISO EN 13501-1 and NFPA 701, and can withstand peak temperatures up to 1,600 °C.

Battery Chemistry Compatibility & Coverage — scored 80/100

> We verified that the company's product documentation focuses heavily on form factors and temperature limits without providing engineering analysis or lists for discrete battery chemistries. Based on primary research, the company has not disclosed compatibility lists, test data across chemistries, or application notes mapping performance to specific lithium-ion battery chemistries like NMC or LFP.

SUPPORTING POINTS

- The company reports providing high-temperature fire blankets, lithium battery storage blankets, and pallet covers designed generally to isolate electric vehicle and lithium-ion battery fires, but does not provide details on specific chemistry compatibility.

Product Portfolio Breadth & Modularity — scored 100/100

> The rating rests on the company's clear and verifiable capability to deliver a modular, pack-and-vehicle-level thermal containment and suppression suite comprised of blankets, aerosol units, and specialized penetrative water lances. Based on primary research, the company has not disclosed cell-level sensors or integrated module-level battery management management system (BMS) safety components within its catalog.

SUPPORTING POINTS

- The company reports a specialized product portfolio centered on its multi-component EV fire response concept, which includes high-temperature resistant fire blankets (Ultimate FI-BL0906, Ultra, and Economy variants), handheld potassium nitrate aerosol units, and water mist applicators/lances (including the patented Water Mist Applicators 126, 138, and 241 for underbody cooling).
- The company reports expansion of this portfolio to include specific smaller-scale and logistics options, such as fire blankets tailored for e-bikes, e-scooters, forklifts, and lithium battery pallet covers.

R&D and Innovation Pipeline — scored 86/100

> The assessment rests on the company's verified commercial release of mechanical and chemical battery fire containment equipment, including penetrating lances and potassium nitrate aerosol units deployed together as a tactical concept. Based on primary research, the company has not disclosed its dedicated R&D expenditure metrics, quantitative internal investment figures, or specific algorithmic software integration with automotive battery management systems.

SUPPORTING POINTS

- The company reports the launch of the Fire Isolator EV Fire Gun (FI-FIREGUN) in February 2026, a professional, insulated firefighting tool designed to pierce an electric vehicle's battery pack housing and flood it internally with water.
- The company reports the implementation of patented water mist applicators (available in sizes 126, 138, and 241) to cool the battery at the source and reduce toxic vapors under its high-temperature vehicle fire containment blankets.

Certifications & Standards Compliance — scored 95/100

> We verified that the company's compliance framework rests heavily on specialized component-level certifications for fire textile propagation and aerosol safety rather than large-scale battery system standards. Based on primary research, the company has not disclosed certifications or test compliance under battery-specific safety standards such as UL 9540A, NFPA 855, or IEC 62619.

SUPPORTING POINTS

- The company reported that all three of its fire blanket models completed testing and achieved the German DIN SPEC 91489 certification in August 2025. Fire Isolator
- The company reports that its Ultimate and Ultra fire blankets possess ISO EN 13501-1 and NFPA 701 certifications, and its accompanying aerosol units hold EN 15276-1 and UL 2775 product listings.

Y — BUSINESS CAPABILITY

Named Customers & Contracts — scored 90/100

> The evaluation rests on the company's confirmed distribution partnership in Italy and its self-reported usage by automotive brands, though specific end-user contracts remain generic. Based on primary research, the company has not disclosed specific customer names, contract values, or finalized procurement agreements with battery manufacturers or electric vehicle OEMs.

SUPPORTING POINTS

- The company announced a strategic partnership with EMME Antincendio to distribute its electric vehicle fire control products across the Italian market.
- The company reports that its fire blankets are used in workshops, EV service locations, and industrial facilities, claiming that almost all major car brands utilize them for lithium-ion battery fire isolation.

Revenue & Growth — scored 73/100

> Based on primary research, the company has not disclosed annual revenue and growth rate from battery fire safety products and services.

SUPPORTING POINTS

- Not available

Geographic Footprint & Channel Network — scored 92/100

> We verified that Fire Isolator BV possesses an extensive global sales footprint backed by a distributed international dealer network spanning multiple continents. Based on primary research, the company has not disclosed the precise number of individual distribution entities or authorized sales partners within its network.

SUPPORTING POINTS

- The company reports maintaining a global dealer network across over 50 countries, including the United States, Canada, Japan, South Korea, India, Australia, and numerous nations across Europe and the Middle East.
- The company reports operating as part of the Heijboer Safety Group, which maintains 4 corporate and office locations across the Netherlands and Germany.

Installed Base & Market Share — scored 96/100

> The assessment rests on the company's self-reported global distribution and deployment data across multiple vehicle logistics and public safety sectors, though these assertions remain unverified by independent census metrics. Based on primary research, the company has not disclosed exact installed unit counts or its specific percentage of market share.

SUPPORTING POINTS

- The company reports its electric vehicle fire containment systems are deployed globally by automotive manufacturers, car carriers, parking operators, and professional fire brigades across North America, Europe, the Middle East, and Asia.
- The company reports that its product concept has been validated through more than 15 live fire tests conducted on hybrid and full electric vehicles.

Funding & Strategic Partnerships — scored 85/100

> We verified that the company's market footprint relies on regional safety group integration and European distribution alliances rather than institutional investment. Based on primary research, the company has not disclosed any external funding rounds, venture capital investment, or strategic alliances with battery manufacturers.

SUPPORTING POINTS

- Fire Isolator BV announced a partnership with EMME Antincendio on September 25, 2024, to distribute its electric vehicle fire control products through the manufacturer's established Italian distribution network.
- The company reports that its core fire isolation concept was co-developed by T-ISS BV, Studsgaard A/S, and Garant Protech, and it operates under the umbrella of the Heijboer Safety Group.

RETRON GmbH RETRON ● Challengers X 88 Y 83 Overall 86

X — PRODUCT CAPABILITY

Fire Suppression Technology & Performance — scored 95/100

> The evaluation rests on the company's verified BAM-GGR 024 compliance and containment fire testing for battery storage and transport enclosures, which demonstrate solid passive protection capabilities. Based on primary research, the company has not disclosed active fire suppression metrics such as precise agent deployment times, active detection speeds, or independent UL 9540A testing data.

SUPPORTING POINTS

- The company's large battery containers have received BAM-GGR 024 certification from the German Federal Institute for Materials Research and Testing (BAM) for transporting critically defective lithium-ion batteries up to 169 kWh, following fire tests demonstrating that external container temperatures do not exceed 100°C and no flames, liquids, or splinters escape during an induced thermal runaway.
- The company reports that its RETRON BOX utilizes fire-retardant cushions to partially filter toxic or corrosive discharging gases and absorb leaked electrolytes, while leveraging a hot-dip galvanized steel design to prevent the escape of sparks and flames.

Battery Chemistry Compatibility & Coverage — scored 85/100

> The rating rests on confirmed regulatory certifications for handling large-scale lithium-ion battery loads up to 169 kWh. Based on primary research, the company has not disclosed clear compatibility lists, application notes, or thermal testing datasets broken down by specific cell chemistries (such as NMC and LFP) or physical form factors (such as cylindrical, prismatic, and pouch cells).

SUPPORTING POINTS

- The company reports that its fireproof safety containers (such as the RETRON BOX, 240, 460, 600, 750, and 4000) are designed generally for the storage, charging, and transport of lithium-ion and hybrid batteries across small electronic devices up to large electric car batteries.
- The company reports that its containers have achieved German Federal Institute for Materials Research and Testing (BAM) approval in accordance with BAM-GGR 024 for transport of critically defective lithium-ion batteries up to 169 kWh.

Product Portfolio Breadth & Modularity — scored 83/100

> We verified that RETRON GmbH offers a scalable product portfolio consisting of distinct sizes of passive thermal containment and safety transport boxes. Based on primary research, the company has not disclosed cell-level or module-level active sensors and pack-level internal suppression systems within its standard catalog.

SUPPORTING POINTS

- The company reports a comprehensive line of fire-resistant safety containers (including the RETRON BOX, 240, 460, 600, 750, and 4000) that serve as customizable or modular storage, charging, and transport mitigation systems for lithium-ion batteries across household, commercial, and large industrial capacities.
- The company reports that its container system is certified by Germany's BAM (Federal Institute for Materials Research and Testing) under standard GGR 024, confirming thermal runaway containment up to 169 KWh for critically defective electric car batteries.

R&D and Innovation Pipeline — scored 86/100

> We verified that the company has proven material-containment capabilities backed by independent German federal certification for large-scale battery thermal runaway events. Based on primary research, the company has not disclosed specific metrics regarding R&D expenditure, patent filings, or advanced electronic detection algorithms integrated into its systems.

SUPPORTING POINTS

- The company reports the development of hot-dip galvanized steel safety containers equipped with non-flammable insulation layers and fire-resistant cushions that filter gases, absorb leaked electrolytes, and limit external temperatures below 100°C during an induced thermal runaway.
- RETRON GmbH completed a fire test and achieved BAM-GGR 024 certification from the German Bundesanstalt für Materialforschung und -prüfung in November 2024, verifying the containment capability of its transport containers for critically defective lithium-ion batteries up to 169 kWh.

Certifications & Standards Compliance — scored 93/100

> The assessment rests on the company's third-party BAM-GGR 024 fire-testing approval and its ADR dangerous goods transport compliance for containment systems. Based on primary research, the company has not disclosed compliance or testing records mapped to international stationary energy storage standards such as UL 9540A, NFPA 855, or IEC 62619.

SUPPORTING POINTS

- The company reports that its storage and transport containers achieved BAM-GGR 024 certification from the German Bundesanstalt für Materialforschung und -prüfung in November 2024, confirming fire containment safety for critically defective lithium-ion batteries up to 169 kWh.
- The company reports that its RETRON BOX and larger safety containers maintain UN approval and ADR compliance for the legally compliant transport of lithium-ion batteries.

Y — BUSINESS CAPABILITY

Named Customers & Contracts — scored 72/100

> We verified that no specific public announcements regarding signed customer names, exact contract values, or direct partnership agreements with specific automotive OEMs or battery manufacturers have been released. Based on primary research, the company has not disclosed named commercial customer contracts or public end-user contract values.

SUPPORTING POINTS

- Not available

Revenue & Growth — scored 94/100

> We verified that the company's operational volumes and qualitative growth descriptions are disclosed in its corporate communications following its restructuring into a legally independent entity in 2023. Based on primary research, the company has not disclosed annual revenue figures, revenue breakdowns, or year-over-year growth percentages specifically derived from its battery fire safety products and services.

SUPPORTING POINTS

- The company reports that its solutions are very well received on the market and demand is increasing, which led to taking its steady growth into account by becoming legally independent as of 1 January 2023.
- The company reports that it takes back over 1,000 tonnes of damaged lithium-ion batteries every year and hires out 1,500 containers, experiencing volumes that continue to steadily grow.

Geographic Footprint & Channel Network — scored 92/100

> The company's geographic evaluation rests on its verified position within its parent organization's international operational footprint spanning over 30 countries, alongside direct evidence of localized container logistics serving third-party clients in Switzerland. Based on primary research, the company has not disclosed the precise quantity of its own dedicated international offices, proprietary distributors, or independent sales partners.

SUPPORTING POINTS

- The company reports its main corporate operations out of Lünen, Germany.
- The company leverages a cross-border logistics network as a subsidiary of the REMONDIS Group, which maintains approximately 1,500 branches and associated entities spanning over 30 countries.

Installed Base & Market Share — scored 69/100

> Based on primary research, the company has not disclosed its exact installed base or market share data.

SUPPORTING POINTS

- Not available

Funding & Strategic Partnerships — scored 88/100

> The rating rests on the company's verified position as an integrated subsidiary within the corporate network of a global recycling and service group rather than an independent startup raising standard venture rounds. Based on primary research, the company has not disclosed external venture capital funding rounds, traditional venture investors, or independent cross-industry joint ventures outside its parent group.

SUPPORTING POINTS

- The company reports that it operates as a subsidiary of REMONDIS Industrie Service GmbH & Co. KG, allowing it to cooperate with partner companies within the REMONDIS Group network.

Fire Cloak Limited Fire Cloak ● Challengers X 89 Y 82 Overall 86

X — PRODUCT CAPABILITY

Fire Suppression Technology & Performance — scored 90/100

> The rating rests on the company's confirmed alignment with established fire resistance and vehicle blanket standards, demonstrating an entry-level thermal isolation and mitigation capability. Based on primary research, the company has not disclosed independent quantitative test data regarding its specific suppression times, interior temperature reduction rates, or performance metrics under formal UL 9540A protocols.

SUPPORTING POINTS

- The company reports that its Fire Cloak electric vehicle fire blankets are certified to DIN SPEC 91489:2024-11, BS476 Parts 6 & 7 (Class 0), and ASTM D6143 for managing electric vehicle and advanced battery fires.
- The company reports that the product fabric is engineered using silica quartz coated with a fire-resistant polymer to resist temperatures exceeding 1,000°C (1,832°F).

Battery Chemistry Compatibility & Coverage — scored 90/100

> The rating rests on confirmed live-testing data involving a production electric vehicle and self-reported thermal resistance handling up to 1,600°C thermal stress. Based on primary research, the company has not disclosed explicit compatibility lists broken down by specific cell chemistries (such as NMC or LFP) or distinct battery form factors (such as cylindrical, prismatic, or pouch).

SUPPORTING POINTS

- The company reports that its Fire Cloak™ Lithium Battery Fire Limitation Blanket is designed broadly for vehicle lithium battery packs and electric vehicles undergoing thermal runaway, managing thermal spikes from 100°C to 1,000°C.
- Public test literature shows the blanket was live-tested at Horiba MIRA over a burning Nissan Leaf, successfully containing the vehicle's thermal runaway event.

Product Portfolio Breadth & Modularity — scored 82/100

> The assessment rests on the company's established, standard-sized product catalog of containment blankets and pallet covers optimized for vehicular and logistics applications. Based on primary research, the company has not disclosed cell-level, module-level, or integrated pack-level sensors and active mechanical suppression systems.

SUPPORTING POINTS

- The company reports a standardized product lineup consisting of the Fire Cloak Lithium Battery Fire Limitation Blanket (4x3m) for smaller lithium risks like e-bikes and workshop replacements, the EV Fire Limitation Blanket (8x6m) for passenger cars, and the XL EV Fire Limitation Blanket (12x9m) for large commercial vehicles.
- The company reports the launch of Fire Cloak Pallet Safe, a specialized high-performance lithium battery fire protection cover designed for logistics hubs and warehouses to isolate high-risk palletized batteries during storage or transit.

R&D and Innovation Pipeline — scored 82/100

> The assessment rests on the verified achievement of international safety standard certification for its commercialized silica quartz and polymer-coated lithium-ion battery fire blankets. Based on primary research, the company has not disclosed specific R&D expenditure figures, active patent filings, or advanced digital integration systems like detection algorithms.

SUPPORTING POINTS

- The company reports developing the Fire Cloak EV Car fire blanket following months of research, development, and testing to isolate and contain thermal runaway fires.
- Prosol UK announced that the Fire Cloak EV Fire Limitation Blanket completed a full testing program and officially achieved certification to the DIN SPEC 91489:2024-11 standard in November 2024.

Certifications & Standards Compliance — scored 100/100

> The rating rests on the company's extensive third-party testing and certification against specialized fire containment blanket specifications such as DIN SPEC 91489:2024-11. Based on primary research, the company has not disclosed certifications for broader stationary storage or industrial battery safety frameworks such as UL 9540A or IEC 62619.

SUPPORTING POINTS

- The company reports that its Fire Cloak™ EV Fire Limitation Blanket has achieved certification to DIN SPEC 91489:2024-11, which tests thermal resistance, deployment performance, and mechanical strength for lithium battery fire containment.
- The company's fire blankets are tested and certified to multiple fire and flame resistance standards including BS 476 Class 0, BS EN 13501-1, NFPA 701, and ASTM D6413.

Y — BUSINESS CAPABILITY

Named Customers & Contracts — scored 89/100

> The assessment rests on a confirmed deployment with an automotive services company to provide mitigation measures for electric vehicle fires. Based on primary research, the company has not disclosed major long-term contracts, transaction values, or formal partnership agreements with automotive electric vehicle OEMs or utility-scale energy storage integrators.

SUPPORTING POINTS

- Prosol UK announced in July 2022 that its Fire Cloak Electric Vehicle Fire Blankets were deployed in sit-u within the electric vehicle health and safety bays of BCA UK.

Revenue & Growth — scored 70/100

> We verified that Fire Cloak is an electric vehicle fire containment blanket brand manufactured and distributed by Prosol UK Sales and Distribution Limited. Based on primary research, the company has not disclosed annual revenue and growth rate from battery fire safety products and services.

SUPPORTING POINTS

- Not available

Geographic Footprint & Channel Network — scored 92/100

> We verified that the company relies on a network of international distribution partners across several continents to support its global sales presence. Based on primary research, the company has not disclosed the exact total number of countries served or specific regional office counts.

SUPPORTING POINTS

- The company reports that its electric vehicle fire blanket is distributed globally, with operations and partnerships spanning the United Kingdom, Taiwan, South Korea, Australia, the United States, South America, and mainland Europe.
- The company has established partnerships with regional entities, including a distribution agreement with HeTone Greenergy in Taiwan and Consolidated Dealers in Canada.

Installed Base & Market Share — scored 71/100

> We verified that the company has deployed its systems to commercial automotive networks like Lithia UK. Based on primary research, the company has not disclosed its exact volume of installed units or its total percentage share of the battery fire safety market.

SUPPORTING POINTS

- Lithia UK has purchased a Fire Cloak vehicle blanket for each of its motor retail operations to manage the risks posed by electric vehicle battery fires.

Funding & Strategic Partnerships — scored 87/100

> We verified that the brand's rating rests on documented distribution alliances and recurring supply agreements within elite motorsport applications. Based on primary research, the company has not disclosed any institutional venture capital rounds, private equity investments, or formal research consortium grants.

SUPPORTING POINTS

- The company announced a sustained partnership with Taiwan-based distributor HeTone Greenergy in July 2026 to supply premium lithium battery fire protection blankets.
- The company reports that its specialized electric vehicle fire safety blankets continue to be selected by Formula E for deployment across the grid for the 2026 season.

Brimstone Fire Protection Brimstone Challengers X 87 Y 83 Overall 85

X — PRODUCT CAPABILITY

Fire Suppression Technology & Performance — scored 93/100

> The assessment rests on the company's verified compliance with UL Standard 5800 for its line of co-developed personal electronic device containment bags. Based on primary research, the company has not disclosed independent third-party thermal runaway suppression times, cooling metrics, or UL 9540A test reports for its electric vehicle blankets and under-chassis mister systems.

SUPPORTING POINTS

- The company reports that its Accenture PED Safety Bag achieved UL Standard 5800 Class 2 and Class 3 certification, co-developed with Accenture, to provide containment of flames, heat, and gas for personal electronic devices.
- The company reports that its EV Fire Suppressor Mister™ delivers a 10 gpm low-flow water mist spray directly under a vehicle chassis, which tests validate is as effective as high-flow models at mitigating thermal runaway while reducing toxic runoff.

Battery Chemistry Compatibility & Coverage — scored 85/100

> The rating rests on confirmed physical containment product availability for varied application profiles, including aviation, drones, and electric vehicles. Based on primary research, the company has not disclosed explicit compatibility lists, performance matrices, or thermal runaway test data across specific lithium-ion battery chemistries like NMC or LFP, nor variations across cylindrical, prismatic, and pouch form factors.

SUPPORTING POINTS

- The company reports testing and containment capabilities across various personal electronic devices, drone batteries, robotic platform cells, and electric vehicle battery modules ranging from 29.6 Wh to 160 Wh capacities.

Product Portfolio Breadth & Modularity — scored 79/100

> We verified that the portfolio provides scalable external suppression and containment options ranging from small portable electronics up to full-sized electric vehicles. Based on primary research, the company has not disclosed cell-level sensors or integrated module- and pack-level internal suppression systems.

SUPPORTING POINTS

- The company reports a product portfolio that spans Brimstone Fire Protection Containment Bags rated for 29.6 Wh to 160 Wh personal electronics, heavy-duty industrial fire blankets up to 16'x16', and vehicle-scale EV Fire Suppressor Blankets and under-carriage Misters.
- The company reports that it designs and builds custom containment solutions for unique lithium-ion battery fire hazards, providing quotes for specialized projects within a few days.

R&D and Innovation Pipeline — scored 86/100

> The evaluation rests on the company's confirmed product engineering capabilities, demonstrated by its UL 5800-rated containment bags co-developed with Accenture and its proprietary pressure relief valves. Based on primary research, the company has not disclosed audited R&D spending figures, a comprehensive formal patent portfolio, or active research pipelines involving advanced detection algorithms and direct integration with battery management systems.

SUPPORTING POINTS

- Brimstone Fire Protection developed the Accenture PED Safety Bags, which are co-developed with Accenture and achieved UL Standard 5800 Class 2 and Class 3 Performance Level 1 ratings for total lithium-ion battery fire and smoke containment.
- The company reports utilising a patent-pending pressure relief valve technology integrated within its lithium-ion fire and smoke containment bags to prevent rupture from over-pressure during thermal runaway.

Certifications & Standards Compliance — scored 93/100

> The company's compliance evaluation rests on its verified product alignment with UL Standard 5800 and FAA flame-resistance materials specifications for aviation electronic device safety containment. Based on primary research, the company has not disclosed compliance data or third-party certifications regarding stationary or large-scale battery storage standards such as UL 9540A, NFPA 855, or IEC 62619.

SUPPORTING POINTS

- The company reports that its Accenture PED Safety Bags, co-developed with Accenture, are certified to UL Standard 5800 (Class 2 and Class 3 Performance Level 1) for fire, smoke, flame, debris, and invisible gas containment.
- The company reports that materials used in its Preventer Li-Ion Fire Containment Bags comply with FAA 14 CFR 25, Appendix F materials standards for fire resistance in aircraft cabin and cargo areas.

Y — BUSINESS CAPABILITY

Named Customers & Contracts — scored 91/100

> The evaluation rests on the company's detailed disclosure of premium aerospace, automotive, and technology clients alongside a co-development partnership with Accenture. Based on primary research, the company has not disclosed specific contract values or official deployment dates for these customer relationships.

SUPPORTING POINTS

- The company reports that its lithium-ion fire protection products are utilized by major global entities, including Mercedes-Benz through its standard service program, NASA for space flight approval/ISS stowage, Qantas, NetJets, Northrop Grumman, Hewlett Packard, General Motors, Google, Tesla, SpaceX, Amazon, and various US military branches.
- The company reports a co-development partnership with Accenture for its UL Standard 5800-rated Class 2 and Class 3 Accenture PED Safety Bags, which are authorized for sale in the United States.

Revenue & Growth — scored 79/100

> Based on primary research, the company has not disclosed annual revenue and growth rate from battery fire safety products and services.

SUPPORTING POINTS

- Not available

Geographic Footprint & Channel Network — scored 89/100

> The evaluation is based on the company's verified primary headquarters in Indiana and its structured global partner network across regions like the United States, Australia, and Mexico. Based on primary research, the company has not disclosed the total number of physical offices or exact count of countries served.

SUPPORTING POINTS

- The company maintains its primary headquarters location in Harlan, Indiana, USA, operating as an Indiana-owned company.
- The company works with a worldwide team of distributors, including designated regional sales partners such as Atlantic Emergency in the United States, Big Red Truck Fire Rescue in Australia, and ECOMAX in Mexico.

Installed Base & Market Share — scored 71/100

> Based on primary research, the company has not disclosed its installed unit counts or exact market share percentages.

SUPPORTING POINTS

- Not available

Funding & Strategic Partnerships — scored 86/100

> The rating rests on the company's recorded strategic supply collaborations with Leidos, NASA, and Mercedes-Benz for its specialized lithium-ion battery fire containment bags and blankets. Based on primary research, the company has not disclosed any institutional venture capital or private equity funding rounds.

SUPPORTING POINTS

- The company reports a successful collaboration announced in September 2019 with Leidos to supply NASA with a lithium-ion battery storage solution for the International Space Station.
- The company reports that its products are utilized as standard service equipment within the Mercedes-Benz service program.

Guard-K Co., Ltd. Guard-K ● Challengers X 90 Y 78 Overall 84

X — PRODUCT CAPABILITY

Fire Suppression Technology & Performance — scored 89/100

> The rating rests on the company's established presence in manufacturing commercial containment covers and blankets designed to mitigate electric vehicle and secondary battery fire hazards. Based on primary research, the company has not disclosed independent, third-party quantitative test metrics (such as UL 9540A data), precise temperature reduction variables, or official fire suppression timelines.

SUPPORTING POINTS

- The company reports that Guard-K firecovers and blankets are engineered to safely contain lithium battery fires, isolate the blaze, prevent oxygen from fueling the reaction, and facilitate the necessary cooling process.

Battery Chemistry Compatibility & Coverage — scored 91/100

> We verified that the company's fire suppression device successfully passed the NTA 8133 standard for lithium-ion battery fire containment. Based on primary research, the company has not disclosed comprehensive compatibility metrics regarding performance variances across specific individual cell chemistries like NMC and LFP or variations across cylindrical, prismatic, and pouch cell form factors.

SUPPORTING POINTS

- The company reports that its "PRIVATE" fire suppression device successfully passed the NTA 8133 lithium-ion battery fire test standard in September 2024.
- The company reports manufacturing specialized lithium battery storage cabinets, fireproof boxes, and eco-friendly fire blankets (firecovers) to isolate and cool lithium battery fires.

Product Portfolio Breadth & Modularity — scored 94/100

> The rating rests on the company's clear range of containment and suppression products across electric vehicle, storage cabinet, and rack-level applications. Based on primary research, the company has not disclosed granular cell-level modular sensors or custom configurations for pack-integrated thermal management systems.

SUPPORTING POINTS

- The company reports a specialized portfolio of lithium-ion fire safety solutions, including diverse types of EV fire blankets (Pro, Carbon, Fibreglass, High Silica), fire storage boxes in dimensions up to 1200x1800x600, storage cabinets with automated fireproof doors closing at 75°C, and dedicated secondary battery collection boxes.
- The company has expanded its application reach into system-level infrastructure, filing patents for purpose-built battery-rack fire suppression systems and automated server-rack fire blankets designed to isolate fires via gravity-driven lowering mechanisms.

R&D and Innovation Pipeline — scored 86/100

> The rating rests on the development and commercial launch of passive and active battery fire safety containment systems, including firecovers and specialized storage cabinets featuring automatic isolation thresholds. Based on primary research, the company has not disclosed its exact R&D spending figures, specific patent filings, or technical details regarding its proprietary lithium-specific fire suppression agents.

SUPPORTING POINTS

- The company reports developing specialized fire safety technologies including natural fibers that can withstand 1,000°C for one hour (up to 1,600°C short-term) and an "Astrosolution" water-based fire-retardant coating designed to prevent static electricity build-up.
- The company's Lithium Battery Storage Cabinet features a specialized monitoring system to track internal conditions and an automatic fireproof door designed to seal off oxygen supply when the temperature reaches 75°C.

Certifications & Standards Compliance — scored 92/100

> Assessment rests on confirmed local and regional test validations, including KFI and TÜV SÜD documentation for its fire blankets and containment items. Based on primary research, the company has not disclosed compliance or testing records for large-scale international energy storage system safety metrics such as UL 9540A, NFPA 855, or IEC 62619.

SUPPORTING POINTS

- The company reports obtaining a TÜV SÜD test report and CE marking (Declaration of Conformity) in August 2024.
- The company reports receiving class B certification from the Korea Fire Institute (KFI) for its lithium-ion battery fire extinguisher.

Y — BUSINESS CAPABILITY

Named Customers & Contracts — scored 89/100

> The evaluation rests on the company's confirmed public business agreement to deploy its specialized fire cover solution for electric vehicle battery fires with a major domestic security and safety services firm. Based on primary research, the company has not disclosed specific commercial contract values, shipment volumes, or named battery manufacturer and automotive OEM purchase orders.

SUPPORTING POINTS

- Guard-K Co., Ltd. announced a business agreement with SK Shieldus in May 2023 to collaborate on safety solution projects related to fire prevention and response, introducing Guard-K's "Fire Cover Pro" inert gas fire extinguishing foam to suppress electric vehicle and battery fires.

Revenue & Growth — scored 73/100

> , Ltd.'s annual revenue, financial breakdowns, or growth percentages specifically tied to its battery fire safety products and services. Based on primary research, the company has not disclosed annual revenue and growth rate from battery fire safety products and services.

SUPPORTING POINTS

- Not available

Geographic Footprint & Channel Network — scored 92/100

> We verified that the company maintains an established presence across South Korea, China, Singapore, and the United States through its regional entity and strategic sales partnerships. Based on primary research, the company has not disclosed the exact number of its international offices, authorized global distributors, or active sales partners.

SUPPORTING POINTS

- The company reports active deployment of its systems in electric vehicle factories, charging stations, and data centers across South Korea, China, and Singapore.
- TradeOn acts as the exclusive U.S. sales partner for Guard-K to market its battery fire safety solutions in American markets.

Installed Base & Market Share — scored 69/100

> Based on primary research, the company has not disclosed the number of units deployed globally or its absolute market share percentage within the battery fire safety segment.

SUPPORTING POINTS

- Not available

Funding & Strategic Partnerships — scored 68/100

> Based on primary research, the company has not disclosed any details regarding its capitalization, investors, or strategic collaboration agreements.

SUPPORTING POINTS

- Not available

Fike Corporation Fike Blue ● Trailblazers X 98 Y 81 Overall 90

X — PRODUCT CAPABILITY

Fire Suppression Technology & Performance — scored 100/100

> The assessment is supported by Fike Corporation's proprietary product performance data and verified UL 9540A testing via CSA. Based on primary research, the company has not disclosed comprehensive, unredacted, or standardized third-party comparative test metrics across a wider variety of commercial battery chemistries.

SUPPORTING POINTS

- The company reports that its Fike Blue thermal management liquid agent was successfully tested by the Canadian Standards Association (CSA) in a UL 9540A test, where it cooled battery modules from hundreds of degrees Celsius to under 100°C in less than 10 minutes.
- The company reports that in internal testing at the Fike Innovation and Testing Center, the application of Fike Blue restricted thermal runaway propagation by isolating the failure to only 6 out of 36 battery cells within the affected module.

Battery Chemistry Compatibility & Coverage — scored 91/100

> The rating reflects the company's established, third-party tested mitigation system and custom testing program that accommodates diverse manufacturer designs, though it lacks an explicit chemistry or form-factor compatibility roster. Based on primary research, the company has not disclosed specific compatibility matrices or test performance data broken down by distinct battery chemistries (such as NMC or LFP) and form factors.

SUPPORTING POINTS

- The company reports that its Battery Hazard Analysis (BHA) program evaluates and designs custom mitigation systems with Fike Blue for a variety of battery types, recognizing that because every battery is unique, a one-size-fits-all thermal management approach is insufficient.

Product Portfolio Breadth & Modularity — scored 100/100

> The assessment is based on the company's established, multi-layered fire safety catalog ranging from rack-level detection to module-level liquid immersion suppression. Based on primary research, the company has not disclosed cell-level active suppression hardware.

SUPPORTING POINTS

- The company reports its Fike Blue thermal management system stops propagating thermal runaway by immersing the affected battery module to absorb exothermic heat, offering a tailored modular discharge via piping networks only into the module where the fire is located.
- The company reports a comprehensive layered portfolio spanning early detection via Li-ion Tamer rack-level off-gas sensors, distributed temperature sensing cables, explosion control venting panels, and external fire suppression utilizing clean agents like FK-5-1-12.

R&D and Innovation Pipeline — scored 98/100

> We verified that the company has a strong verifiable capability in battery fire safety, supported by specialized large-scale testing infrastructure and a proprietary agent validated via third-party UL 9540A testing. Based on primary research, the company has not disclosed its exact annual R&D spending figures or integration roadmaps for advanced internal detection algorithms. Fike Corporation If you would like to expand this analysis, please let me know if we should evaluate Fike Corporation on Geographic Market Penetration, Manufacturing & Scalability, or Regulatory Compliance & Certifications.

SUPPORTING POINTS

- The company reports the development and November 2023 announcement of Fike Blue, a patented, third-party tested thermal management fluid and suppression system designed to immerse battery modules and halt cascading thermal runaway.
- The company announced in September 2025 the expansion of its R&D operations into a new 42,000-square-foot Fike Labs HQ facility in Blue Springs, Missouri, which features four large-scale battery fire test cells explicitly dedicated to thermal runaway testing and product development.

Certifications & Standards Compliance — scored 100/100

> The rating rests on the company's verified third-party testing via CSA Group under the UL 9540A test standard for its thermal runaway suppression agent. Based on primary research, the company has not disclosed specific product-level certifications or formal component listings under NFPA 855 or IEC 62619.

SUPPORTING POINTS

- Fike Corporation reports that its Fike Blue thermal management agent has undergone UL 9540A testing—validated both internally and by CSA Group—to prove its capability in stopping cascading thermal runaway within lithium-ion battery modules.
- Fike Corporation provides a Battery Hazard Analysis (BHA) program designed to empirically test battery modules and design fire protection setups that ensure client assemblies pass UL 9540A certification at a Nationally Recognized Testing Laboratory (NRTL).

Y — BUSINESS CAPABILITY

Named Customers & Contracts — scored 89/100

> We verified that the company has active fire protection deployments for vehicle transport garages and an educational institution. Based on primary research, the company has not disclosed the names or contract values of commercial battery manufacturers, electric vehicle OEMs, or energy storage integrators.

SUPPORTING POINTS

- Fike Corporation collaborated with an unnamed major shipping company to design and supply early warning fire detection systems for roll-on/roll-off vehicle garages storing electric vehicles.
- Fike Corporation announced a partnership in May 2024 with Heriot-Watt University Dubai to protect its rooftop battery energy storage system using Fike Blue and introduce an energy storage safety curriculum.

Revenue & Growth — scored 73/100

> We verified that Fike Corporation actively markets thermal management and suppression systems under its Fike Blue product line to mitigate thermal runaway in battery energy storage systems. Based on primary research, the company has not disclosed annual revenue or growth rates specifically derived from its battery fire safety products and services.

SUPPORTING POINTS

- Not available

Geographic Footprint & Channel Network — scored 89/100

> The evaluation rests on our verification of the company's global corporate infrastructure consisting of manufacturing plants across five countries and regional offices, paired with a reported network of hundreds of international partners. Based on primary research, the company has not disclosed the specific count of distributors and countries dedicated exclusively to its battery fire safety solutions.

SUPPORTING POINTS

- The company reports operating manufacturing facilities in the United States, Belgium, Wales, Canada, and India, with additional sales and service offices located throughout North and South America, Europe, Asia, and the Middle East.
- The company reports that its fire protection systems are represented globally by a distribution network of hundreds of trusted partners.

Installed Base & Market Share — scored 69/100

> Based on primary research, the company has not disclosed its installed unit counts or market share figures.

SUPPORTING POINTS

- Not available

Funding & Strategic Partnerships — scored 85/100

> We verified that the company has established valid technical and academic partnerships to deploy its thermal runaway suppression and detection solutions. Based on primary research, the company has not disclosed any external funding rounds or investment capital raised.

SUPPORTING POINTS

- Fike Corporation announced a strategic partnership with Heriot-Watt University Dubai in May 2024 to protect its rooftop battery energy storage system from lithium battery fires using the Fike Blue suppression system and to implement an advanced curriculum focused on cascading thermal runaway safety.
- Fike Corporation maintains a technology partnership with Fire Rover, highlighted in August 2025, which integrates Fike IR3 Flame Detectors into Fire Rover's remote firefighting solution to rapidly detect and suppress industrial waste facility fires ignited by lithium battery-powered devices.

Bridgehill AS Bridgehill ● Trailblazers X 92 Y 84 Overall 88

X — PRODUCT CAPABILITY

Fire Suppression Technology & Performance — scored 98/100

> The rating rests on third-party ad hoc and independent framework testing that confirms clear thermal isolation performance and rapid temperature reduction capabilities for lithium-ion battery fires. Based on primary research, the company has not disclosed comprehensive standardized UL 9540A testing data metrics.

SUPPORTING POINTS

- Independent ad hoc testing of the Bridgehill AS Car Fire Blanket by Centro Zaragoza on a burning Nissan Leaf showed that the fire blanket dropped the vehicle's temperature from 1,976 °F to 676 °F in under 30 seconds.
- The company reports that its Car Pro X fire blanket has been independently tested by Kiwa MPA Dresden GmbH in accordance with the DIN SPEC 91489 standardized framework for electric vehicle fire blankets.

Battery Chemistry Compatibility & Coverage — scored 91/100

> The rating rests on confirmation that the company markets fire blankets specifically tailored for diverse small-scale and electric vehicle lithium-ion battery applications, with documentation acknowledging distinct behavior between NMC and LFP chemistries. Based on primary research, the company has not disclosed a granular compatibility list mapping product efficacy metrics to specific cell form factors like cylindrical, pouch, or prismatic types.

SUPPORTING POINTS

- The company reports that its Lithium Fire Blanket is designed for direct application on lithium battery packs to isolate exposed or open lithium-ion battery fires, specifically citing applications for e-bikes, e-scooters, laptops, power tools, and open battery modules.
- A technical assessment authored by the company's founder in 2026 notes evaluating suppression methods against variations in battery chemistry, specifically referencing NMC, NCA, and LFP fire behavior and gas production.

Product Portfolio Breadth & Modularity — scored 89/100

> The rating rests on the company's clear verifiable capability in manufacturing external containment and isolation blankets optimized for different vehicle, industrial, and consumer device scales. Based on primary research, the company has not disclosed cell-level hardware sensors or integrated module- and pack-level active suppression systems.

SUPPORTING POINTS

- The company reports a specialized battery fire safety lineup centered on extreme high-heat isolation blankets, including the Lithium Fire Blanket (engineered to cover open battery packs, e-mobility devices, and modules) and the URBAN Fire Blanket (available in wall-mounted and portable formats for small devices).
- The company reports its Extreme Series fire containment blankets are scalable through multiple size options, ranging from small 1.5x1.5 m formats up to large 7x7 m formats customized for forklifts and industrial warehouse settings.

R&D and Innovation Pipeline — scored 86/100

> The rating rests on the company's clear and verifiable track record of material science innovation and recent product launches tailored directly to portable, automotive, and industrial lithium-ion battery containment. Based on primary research, the company has not disclosed specific R&D spending figures or an active patent pipeline.

SUPPORTING POINTS

- The company reports the development and commercial launch of specialized containment solutions, including the Lithium Fire Blanket for open battery packs, the URBAN Fire Blanket for portable lithium-ion devices, and the FireBlock Membrane for rooftop solar installations launching at Intersolar 2026.
- The company reports that its material science R&D has yielded the proprietary BridgeTech Polymer Coating, a fabric compound utilizing pyroxene or silica fibers designed to withstand extreme thermal runaway temperatures up to 1,600°C without requiring water or chemicals.

Certifications & Standards Compliance — scored 97/100

> We verified the company's compliance metrics for portable thermal containment systems, which align with specialized textile and vehicle fire safety frameworks rather than stationary equipment. Based on primary research, the company has not disclosed certifications for asset-level or stationary battery system standards such as UL 9540A, NFPA 855, or IEC 62619.

SUPPORTING POINTS

- The company reports compliance with the DIN SPEC 91489:2024-11 automotive fire blanket standard for its Car Pro X vehicle fire blanket, which was independently tested to verify containment safety.
- The company reports obtaining NFPA 701-2019 certification for its main fire blanket lines, alongside independent performance testing under EN 13501-1 (Class A1), EN ISO 1182, EN ISO 1716, and ASTM D257.

Y — BUSINESS CAPABILITY

Named Customers & Contracts — scored 91/100

> The assessment rests on confirmed distribution partnerships with logistics equipment manufacturers and public sector procurement orders for vehicle containment products. Based on primary research, the company has not disclosed direct customer contracts, volume commitments, or integration agreements with electric vehicle OEMs or battery manufacturers.

SUPPORTING POINTS

- Bridgehill announced a global partnership agreement with Jungheinrich to distribute its forklift and pallet truck fire blankets across Europe, LATAM, and APAC.
- A public purchase order contract (36C24124P0417) was awarded by the US Department of Veterans Affairs on April 15, 2024, for Bridgehill Car Fire Blankets.

Revenue & Growth — scored 70/100

> Based on primary research, the company has not disclosed annual revenue and growth rate from battery fire safety products and services.

SUPPORTING POINTS

- Not available

Geographic Footprint & Channel Network — scored 100/100

> The rating rests on the company's clear and documented international distribution presence across major regions via regional partnerships and localized safety networks. Based on primary research, the company has not disclosed the exact number of global physical office locations or its complete roster of international sales partners.

SUPPORTING POINTS

- The company reports a primary focus on Europe and North America, with its products delivered globally, including usage by 70% of Norwegian fire departments and over 100 German fire departments.
- The company reports a global distribution partnership with Jungheinrich across Europe, Latin America (LATAM), and Asia-Pacific (APAC), alongside regional authorized distributors like CheckFire in the UK, PT Rescue in Australia and New Zealand, and Unique Fire in Malaysia.

Installed Base & Market Share — scored 69/100

> Based on primary research, the company has not disclosed its precise deployed installed base or market share figures within the battery fire safety sector.

SUPPORTING POINTS

- Not available

Funding & Strategic Partnerships — scored 88/100

> The assessment rests on verified commercial alliances with international automotive distribution networks and material handling manufacturers to deploy lithium-ion containment systems. Based on primary research, the company has not disclosed external investment rounds, institutional venture capital backing, or formal joint ventures.

SUPPORTING POINTS

- A global strategic partnership with Jungheinrich to integrate Bridgehill fire blankets into their distribution network for lithium-powered forklift fire containment across Europe, LATAM, and APAC.
- A collaboration agreement with Harald A. Møller to equip Volkswagen, Audi, Škoda, SEAT, and CUPRA automotive dealers with specialized electric vehicle fire blankets.

CellBlock FCS LLC CellBlockEX ● Trailblazers X 93 Y 82 Overall 88

X — PRODUCT CAPABILITY

Fire Suppression Technology & Performance — scored 93/100

> The assessment rests on extensive third-party laboratory evaluations by entities like Underwriters Laboratories and CSA Group, alongside documented rapid suppression metrics for the proprietary CellBlockEX granulate. Based on primary research, the company has not disclosed formalized, universally standardized metrics matching the specific UL 9540A system certification protocol.

SUPPORTING POINTS

- The company reports that its passive fire suppression system using CellBlockEX mineral-based granulates achieves rapid containment from spark to suppression in approximately three seconds during battery thermal runaway events.
- The company's technology has undergone testing by third-party laboratories including Underwriters Laboratories, CSA Group, and Stress Engineering Group, validating its ability to halt propagation and protect adjacent shelves.

Battery Chemistry Compatibility & Coverage — scored 94/100

> We verified that the company's hardware intake protocols and product materials account directly for standard lithium-ion form factors and major chemistries such as NMC and LFP. Based on primary research, the company has not disclosed explicit third-party verified compatibility charts or comparative test data sheets detailing individual performance variances across all listed chemistries.

SUPPORTING POINTS

- The company reports utilizing a standardized Battery Specs Worksheet to evaluate client needs based on cell style (cylindrical, prismatic, pouch, or other) and cell type/chemistry (LFP, NMC, NCA, LMO, LCO, LTO, or other).
- The company reports performing safety testing across diverse configurations, including a completed 25 kWh thermal runaway propagation test of its SuperMax case conducted in partnership with an automotive OEM.

Product Portfolio Breadth & Modularity — scored 89/100

> The rating rests on the company's verified testing of its modular battery storage cabinets and charging racks through third-party entities like CSA Group to halt thermal runaway propagation across adjacent bays. Based on primary research, the company has not disclosed cell-level hardware sensors or active internal electronic monitoring integrations within the core modules themselves, relying instead on passive physical suppression and external exhaust monitoring attachments.

SUPPORTING POINTS

- The company reports a comprehensive line of products spanning cell-level/device-level containment up to heavy EV applications, including CellBlockEX dry loose-fill suppression media, personal electronic device fire-suppression pillows, micro-mobility modular charging racks, and larger transportation units such as the SuperMax Case designed to contain up to 25 kWh of electric vehicle battery energy.
- The company reports scalable configurations and customizable options for its product lines, allowing engineering collaboration to tailor its transportation and storage cases to exact customer specifications from small power cells to large machinery batteries.

R&D and Innovation Pipeline — scored 96/100

> The evaluation is based on the company's verified patent updates and successful launch of passive fire containment and infrastructure charging systems that utilize its proprietary aggregate media to manage thermal runaway events. Based on primary research, the company has not disclosed its total R&D spending figures or active development pipelines for software-driven advanced detection algorithms integrated with digital battery management systems.

SUPPORTING POINTS

- CellBlock FCS LLC has established a patented portfolio for battery safety containment, including U.S. Patent 11,559,709 published on October 26, 2023, which outlines a shipping container incorporating an interior liner with apertures to isolate and extinguish lithium-ion battery fires using its passive CellBlockEX dry loose-fill mineral aggregate.
- The company expanded its innovation pipeline by launching the Charge Safe Battery Rack, which secured regulatory approval from the FDNY in March 2023 for e-mobility charging infrastructure by employing automated gravity deployment of its suppression media within multi-bay shelves.

Certifications & Standards Compliance — scored 95/100

> The evaluation is based on the company's verified municipal regulatory approval from the FDNY and extensive aviation-oriented burn certifications aligning with 14 CFR standards. Based on primary research, the company has not disclosed formal third-party certification listings under stationary energy storage standards such as UL 9540A, NFPA 855, or IEC 62619 for its industrial containment systems.

SUPPORTING POINTS

- CellBlock FCS LLC announced in March 2023 that its modular Charge Safe Battery Rack obtained regulatory approval by the FDNY for use in charging lithium-ion batteries.
- The company reports that its LIBIK containment products are engineered to meet the ANSI/CAN/UL 5800 standard, and it holds specialized packaging permits from the U.S. DOT (such as SP 20910, SP 20549, and SP 21442) for shipping lithium-ion batteries.

Y — BUSINESS CAPABILITY

Named Customers & Contracts — scored 91/100

> The rating rests on confirmed distributor partnerships and commercial product selection by a regional airline, alongside several modest federal purchase contracts. Based on primary research, the company has not disclosed the monetary value of its commercial contracts or names of automotive EV OEM and large-scale energy storage integration customers.

SUPPORTING POINTS

- CellBlock FCS LLC partnered with Call2Recycle in an agreement announced in January 2020, making Call2Recycle the official distributor of CellBlockEX fire extinguishing media and collaborative developer of the OneDrum mixed battery recycling solution.
- Avelo Airlines selected the company's CellBlock Lithium-ion Battery Incident Kit (LIBIK) for its onboard safety program in a selection announced in August 2021.

Revenue & Growth — scored 73/100

> Based on primary research, the company has not disclosed annual revenue and growth rate from battery fire safety products and services.

SUPPORTING POINTS

- Not available

Geographic Footprint & Channel Network — scored 92/100

> The rating rests on the company's established distribution footprint across the United States, Australia, and parts of Europe through verified commercial partnerships. Based on primary research, the company has not disclosed the exact total number of countries served or a comprehensive list of its international dealer networks.

SUPPORTING POINTS

- The company reports offering worldwide distribution of its products and maintains its main operational facility in Standish, Maine, alongside a corporate location in Boston, Massachusetts.
- The company reports active regional distribution and training partnerships with Call2Recycle as its exclusive U.S. consumer battery recycling partner, Flame Aviation in the Netherlands for airline cabin simulator training, Ecobatt in Australia, and domestic public safety networks like MES Fire and SafeWare.

Installed Base & Market Share — scored 69/100

> Based on primary research, the company has not disclosed its total installed base or battery fire safety market share data.

SUPPORTING POINTS

- Not available

Funding & Strategic Partnerships — scored 86/100

> The evaluation rests on the company's confirmed commercial distribution and logistics partnerships within the recycling and fire safety sectors. Based on primary research, the company has not disclosed any venture funding rounds, institutional investment data, or direct joint ventures with advanced battery manufacturers.

SUPPORTING POINTS

- CellBlock FCS LLC partnered with Call2Recycle in January 2020 to expand battery fire suppression logistics, designating Call2Recycle as the official distributor of CellBlockEX fire extinguishing media.
- CellBlock FCS LLC partnered with EcoBatt Safety, which acts as the company's exclusive distributor of battery fire containment products across Australasia.

AVL List GmbH AVL Stingray ● Trailblazers X 94 Y 83 Overall 88

X — PRODUCT CAPABILITY

Fire Suppression Technology & Performance — scored 91/100

> The rating rests on the company's commercialized AVL Stingray series, which utilize specialized mechanical penetration mechanisms to deliver localized cooling agents directly inside battery housings during thermal events. Based on primary research, the company has not disclosed specific fire suppression times, quantified temperature reduction metrics, or independent test compliance records such as UL 9540A results.

SUPPORTING POINTS

- The company reports the development of the AVL Stingray One, a mobile battery fire suppression system that utilizes a patented nozzle and penetration needle to puncture the vehicle battery housing and introduce cooling water directly inside the pack to halt thermal runaway.
- The company reports the development of the AVL Stingray Testbed, a fire suppression system for test environments that deploys a localized punch unit to penetrate the battery enclosure with an installation time of approximately two minutes to deliver targeted cooling directly at the source of a thermal event.

Battery Chemistry Compatibility & Coverage — scored 98/100

> We verified that the company has a well-established capability to evaluate battery fire safety and thermal runaway dynamics across all major form factors and standard lithium-ion chemistries through its commercial testbeds and simulation software releases. Based on primary research, the company has not disclosed a standardized, public compatibility index or matrix mapping its safety systems against next-generation advanced solid-state battery chemical variations.

SUPPORTING POINTS

- AVL List GmbH reports that its AVL FIRE™ M simulation software features thermal runaway chemical kinetics coupled with fluid and particle transport models to evaluate cell-to-cell failure propagation across diverse cell chemistries, and the company actively benchmarks thermal propagation differences between Lithium Iron Phosphate (LFP) and Nickel Manganese Cobalt (NMC) chemistries.
- The company's AVL Battery Cell TS™ testing platform and cell fixture shelves are explicitly designed to accommodate and validate multiple battery cell form factors, including cylindrical, prismatic, and pouch formats.

Product Portfolio Breadth & Modularity — scored 89/100

> The rating rests on the company’s specialized physical suppression hardware (Stingray series) and high-fidelity CFD simulation tools used for scaling safety workflows from the cell to the pack level. Based on primary research, the company has not disclosed a commercial product catalog of modular, onboard fire safety containment or suppression systems intended for series-production vehicles outside of testbed and simulation environments.

SUPPORTING POINTS

- The company reports the AVL Stingray™ Testbed, which is a targeted fire suppression and containment system designed for high-voltage electric vehicle battery and prototype testing inside climatic chambers, utilizing a punch unit that penetrates the pack to inject water or liquid nitrogen coolants.
- The company reports its AVL FIRE™ M 3D CFD simulation software suite, which features scalable battery thermal safety tools modeling cell heat release, gas venting, particle ejection, and cell-to-cell pack propagation to design custom physical safety configurations.

R&D and Innovation Pipeline — scored 99/100

> We verified that the company possesses a strong innovation pipeline backed by active hardware product lines, software workflows, and published patents focused on early-stage thermal runaway containment and suppression. Based on primary research, the company has not disclosed the exact financial value of its annual R&D budget allocated specifically toward battery fire safety technologies.

SUPPORTING POINTS

- The company reports the development and commercialization of the AVL Stingray One and AVL Stingray Testbed fire suppression systems, which utilize a patented penetration mechanism to inject coolant directly inside the battery housing at the source of a thermal runaway event.
- The company actively updates its specialized battery engineering toolkits, such as the 2025 R2 release of its AVL FIRE M simulation software, which incorporates coupled chemical kinetics and fluid transport models to simulate cell heat release, gas venting, and particle ejection.

Certifications & Standards Compliance — scored 95/100

> The rating rests on the company's clear and verifiable engineering capabilities in providing simulation frameworks, safety chambers, and localized testbed fire suppression systems that support third-party compliance testing. Based on primary research, the company has not disclosed specific independent listings or product certificates under UL 9540A, NFPA 855, or IEC 62619 for its thermal management hardware.

SUPPORTING POINTS

- The company reports developing a "No Flame-Out" battery pack design that combines multi-scale simulation and abuse testing to help manufacturers achieve containment and align with evolving global safety regulations like China's strict GB 38031-2025 mandate.
- The company reports offering the AVL Stingray™ Testbed, a purpose-built high-voltage battery testbed integrated with automated fire suppression and direct-penetration floor cooling designed to secure critical infrastructure during thermal runaway testing.

Y — BUSINESS CAPABILITY

Named Customers & Contracts — scored 91/100

> The assessment rests on confirmed laboratory testing partnerships and distribution networks for specialized battery fire mitigation hardware. Based on primary research, the company has not disclosed specific procurement contract values or the names of automotive electric vehicle OEMs and battery manufacturing customers integrating these suppression systems.

SUPPORTING POINTS

- A strategic partnership with Green Testing Lab was entered into to support the development, homologation, and safety testing of advanced lithium-ion battery systems.
- Commercial distribution agreements were established for the AVL Stingray One battery fire suppression system, appointing Magirus Lohr across several European regions and Total Safety Solutions for the rest of the world.

Revenue & Growth — scored 75/100

> We verified that the company reports overall corporate performance figures, including 1.83 billion Euros for 2025, but does not provide financial segmentation for its battery fire safety or thermal management products such as the AVL Stingray series. Based on primary research, the company has not disclosed the annual revenue or growth rate derived specifically from battery fire safety products and services.

SUPPORTING POINTS

- The company reports a total corporate revenue of 1.83 billion Euros for 2025.

Geographic Footprint & Channel Network — scored 89/100

> The assessment rests on the company's extensive global corporate footprint and engineering hubs across major automotive regions capable of deploying battery containment and suppression testbeds. Based on primary research, the company has not disclosed the number of third-party distributors or sales partners dedicated exclusively to its battery fire safety product lines.

SUPPORTING POINTS

- The company reports a global affiliate network across over 30 countries and operates 45 to 50 technology and engineering centers worldwide to support its mobility engineering, simulation, and testing solutions, including the AVL Stingray Testbed targeted fire suppression system.
- The company maintains specialized facilities and tech centers in key regional markets including the United States, Germany, South Korea, China, and India to deliver its advanced AVL Battery Testing Solutions and safety chambers.

Installed Base & Market Share — scored 65/100

> Based on primary research, the company has not disclosed its installed base numbers or market share data within the battery fire safety industry.

SUPPORTING POINTS

- Not available

Funding & Strategic Partnerships — scored 94/100

> The assessment rests on the company's active strategic alliances with battery manufacturers and academic research institutions to validate thermal runaway mitigation workflows and next-generation battery safety. Based on primary research, the company has not disclosed specific private funding rounds or outside investments raised exclusively for its battery fire safety product segments.

SUPPORTING POINTS

- Blue Solutions and AVL List GmbH announced the successful completion of a joint safety evaluation program in May 2026 to anticipate the safety and thermal runaway behaviors of solid-state battery cells under demanding operational conditions.
- AVL List GmbH announced a partnership with the CMT – Clean Mobility & Thermofluids Institute at Universitat Politècnica de València in March 2026 to drive the TRACY project, an R&D initiative funded under Spain's Public-Private Collaboration Program focused on understanding and mitigating thermal runaway in lithium-ion batteries.

VLITEX GmbH VLITEX ● Trailblazers X 93 Y 79 Overall 86

X — PRODUCT CAPABILITY

Fire Suppression Technology & Performance — scored 98/100

> The rating is supported by independent material and containment validation from the Saxon Textile Research Institute and the Fraunhofer Institute confirming thermal stability up to 750 °C and containment under simulated thermal runaway conditions. Based on primary research, the company has not disclosed standard automated fire detection integrations, exact suppression time duration windows, or formalized UL 9540A testing metrics.

SUPPORTING POINTS

- Independent testing conducted by the Saxon Textile Research Institute (STFI) verified that the VLITEX E-AUTO fire blanket tissue samples remained structurally stable and flexible after heat treatment at 750 °C for 15 and 30 minutes, outperforming competing products that broke during post-heating buckling tests.
- In an independent product containment test conducted by the Fraunhofer Institute based on VDMA 24994 criteria, the VLITEX Battery-Safe Rack L successfully prevented flame escape, safely held ejected debris, and prevented thermal propagation to adjacent batteries after a 750 Wh lithium-ion battery was intentionally ignited.

Battery Chemistry Compatibility & Coverage — scored 93/100

> The rating rests on the company's confirmed testing of its containment products across varying battery capacities and physical formats (pouch, prismatic, and small electronics cells). Based on primary research, the company has not disclosed a comprehensive matrix detailing exact chemical composition compatibility limits (such as distinct NMC versus LFP performance certifications).

SUPPORTING POINTS

- The company reports fire containment testing conducted by the Fraunhofer Institute on its Battery-Safe Rack L using a total setup of 2 x 750 Wh and 4 x 144 Wh lithium-ion batteries.
- The company reports successful containment testing conducted by ACTS Advanced Car Technology Systems GmbH on its fire blanket using five prismatic battery cells of 150 Ah each, reaching peak internal temperatures up to 1,300 °C.

Product Portfolio Breadth & Modularity — scored 96/100

> The assessment rests on the company's clear range of modular containment fabrics, vehicle fire blankets, and adjustable storage systems developed specifically for varying application scales from small electronics up to industrial battery modules. Based on primary research, the company has not disclosed cell-level sensors or built-in, pack-level active chemical suppression hardware.

SUPPORTING POINTS

- The company reports a comprehensive product line of passive textile fire isolation and containment solutions, including lithium-ion battery fire blankets tailored for electric vehicles (AUTO, E-AUTO, Premium Plus), electric forklift trucks, e-scooters, and open or closed battery modules.
- The company reports a portfolio of modular fire barrier bags, battery safety boxes with adjustable internal dividers for multi-size storage, laptop/smartphone fire protection cases, and a modular "battery safe shelf" storage rack system evaluated by the Fraunhofer Institute.

R&D and Innovation Pipeline — scored 82/100

> We verified that the company's innovation efforts rest on expanding its technical textiles into functional battery fire containment and mitigation systems, as demonstrated by product launches like the EV Fire Blanket Premium Plus and wireless temperature tracking sensors. Based on primary research, the company has not disclosed its specific R&D spending figures or patent registrations.

SUPPORTING POINTS

- The company reports the launch of the EV Fire Blanket Premium Plus, which integrates a controlled degassing system with an integrated cable pull system to enhance safety during electric vehicle battery fires.
- The company reports the development of the wireless Temperature Tracker, a magnetic IP67-rated temperature sensor designed to monitor lithium-ion battery temperatures and transmit data via a mobile connection.

Certifications & Standards Compliance — scored 97/100

> We verified that the company possesses clear capability in European textile fire suppression metrics through confirmed third-party testing by European research institutes and early alignment with the regional DIN SPEC 91489 standard. Based on primary research, the company has not disclosed compliance or test reports for utility-scale stationary energy storage standards such as UL 9540A, NFPA 855, or IEC 62619.

SUPPORTING POINTS

- The company reports that its VLITEX PREMIUM fire blankets for electric vehicles are officially certified under the DIN SPEC 91489 standard, which establishes criteria for fire containment blankets handling lithium-ion battery fires.
- The company reports third-party testing confirmations for its material properties, including temperature testing by KIWA up to 1,000 °C, fire module testing by Magna Steyr up to 1,300 °C, a fire resistance test by the Fraunhofer Institute based on VDMA 24994, and cut resistance according to ISO 13997:1999-08 verified by the Sächsisches Textilforschungsinstitut (STFI).

Y — BUSINESS CAPABILITY

Named Customers & Contracts — scored 82/100

> The rating rests on the confirmation of a commercial distribution partnership for the company's battery fire containment textiles. Based on primary research, the company has not disclosed specific names, contract values, or deployment announcements for battery manufacturers, automotive OEMs, or energy storage integrators.

SUPPORTING POINTS

- VLITEX GmbH announced a distribution partnership with Aspli Safety Specialists in January 2025 to offer its range of lithium-ion battery fire blankets and protection products.

Revenue & Growth — scored 73/100

> Based on primary research, the company has not disclosed annual revenue breakdowns or year-over-year growth percentages for its fire safety products and services.

SUPPORTING POINTS

- Not available

Geographic Footprint & Channel Network — scored 92/100

> The rating rests on the company's established presence in Germany and Northern Europe, alongside confirmed international distribution channels in markets like Australia and the UK. Based on primary research, the company has not disclosed the total number of global offices, international countries served, or a comprehensive directory of its global distribution partners.

SUPPORTING POINTS

- The company reports its main corporate headquarters and manufacturing operations are located in Bayreuth, Germany, with dedicated international sales outreach and geographic operations managed via a regional subsidiary entity, VLITEX Nordic AB, established in 2022.
- The company has expanded its distribution network outside Germany through distribution partnerships, notably securing Fire Response as its authorized supplier in Australia and announcing a distribution alliance with Aspli Safety Specialists in January 2025.

Installed Base & Market Share — scored 69/100

> Based on primary research, the company has not disclosed its precise installed unit metrics or market share data within the battery fire safety segment.

SUPPORTING POINTS

- Not available

Funding & Strategic Partnerships — scored 81/100

> We verified that the company's market activity is supported by international distribution partnerships, such as its alliance with Aspli in the United Kingdom. Based on primary research, the company has not disclosed any venture capital funding rounds, institutional investors, or joint ventures with battery manufacturers.

SUPPORTING POINTS

- VLITEX GmbH announced a strategic sales partnership with UK-based safety equipment supplier Aspli in January 2025 to distribute its DIN SPEC-certified lithium-ion battery fire blankets and containment products.

LithiumSafe B.V. LithiumSafe ● Evolving Players X 87 Y 76 Overall 82

X — PRODUCT CAPABILITY

Fire Suppression Technology & Performance — scored 90/100

> We verified that the company's capability assessment rests on self-reported test results for external flame containment using a 7 kWh battery setup and descriptions of its porous gas-filtering filter mechanism. Based on primary research, the company has not disclosed independent third-party fire suppression or thermal management certifications such as UL 9540A testing data, precise suppression times, or specific internal temperature reduction metrics.

SUPPORTING POINTS

- The company reports validating its LithiumSafe Battery Storage Box via a 7 kWh battery thermal runaway test in its model XL, which successfully prevented external flame propagation.
- The company reports that its proprietary porous gas-filtering lid system is designed to relieve high volumes of flammable gas while blocking flames and retaining a large portion of harmful toxic gases within its porous structure during a thermal runaway event.

Battery Chemistry Compatibility & Coverage — scored 90/100

> The assessment rests on the company's documented testing across multiple unspecified battery chemistries and small-to-midsize battery form factors for its bags and containment boxes. Based on primary research, the company has not disclosed a comprehensive explicit list of compatible lithium-ion chemistries (such as NMC versus LFP specific validation benchmarks) or structured data regarding performance across specific cylindrical, pouch, and prismatic cell form factors.

SUPPORTING POINTS

- The company reports that its LithiumSafe™ Battery Box has been tested in over 20 official trials across varying capacities and chemistries to handle thermal runaway.
- The company reports that its containment products are designed to accommodate multiple device and battery formats including laptops, tablets, smartphones, power banks, and spare cells.

Product Portfolio Breadth & Modularity — scored 83/100

> The rating rests on the company's demonstrated range of passive containment and isolation equipment designed for portable electronics, e-bikes, and small modules, including modular safety add-ons like filtration and cooling. Based on primary research, the company has not disclosed cell-level active sensors or full-scale active fire suppression systems integrated into fixed utility-scale pack architectures.

SUPPORTING POINTS

- The company reports a specialized product portfolio focused on containment and isolation systems, including flexible fire containment bags (such as laptop, tablet, and e-bike battery bags), secondary smoke containment sleeves, fireproof battery boxes, and wall-mounted kits.
- The company reports that its rigid enclosures and containment boxes can be customized on request and configured with optional add-ons, including active thermostat-controlled cooling ventilators, internal charging cable entries, and integrated filtration systems to handle thermal runaway events.

R&D and Innovation Pipeline — scored 79/100

> We verified that the company has established a functional line of passive containment products anchored by its patented e-bike battery bag technology. Based on primary research, the company has not disclosed its specific R&D expenditure figures or a formal pipeline for advanced detection algorithms and active suppression integration.

SUPPORTING POINTS

- LithiumSafe reports a patented LithiumSafe 750 Bag designed for 750 Wh e-bike battery fire containment at 100% charge.
- The company reports the development and testing of its LithiumSafe™ Battery Box, which handles thermal runaway incidents up to 7000 Wh and temperatures reaching up to 1400°C.

Certifications & Standards Compliance — scored 95/100

> The assessment rests on confirmed third-party thermal runaway testing from Peutz Laboratories alongside compliance documentation for aviation flammability frameworks including CS 25.853 and UL5800. Based on primary research, the company has not disclosed compliance certifications or testing data related to stationary or industrial energy storage system standards such as UL 9540A, NFPA 855, or IEC 62619.

SUPPORTING POINTS

- The company reports that its LithiumSafe Battery Bag is certified according to both EASA and FAA flammability standard CS 25.853.
- The company reports that its Next Generation LithiumSafe PED Kit is engineered to meet the requirements of the UL5800 aviation personal electronic device fire containment standard, and its e-bike battery bags hold third-party thermal runaway fire test report C 2423-4E-RA-002 from Peutz Laboratories.

Y — BUSINESS CAPABILITY

Named Customers & Contracts — scored 65/100

> V. in the battery fire safety market. Based on primary research, the company has not disclosed customer names, contract values, or formal partnership agreements with OEMs or energy storage integrators.

SUPPORTING POINTS

- Not available

Revenue & Growth — scored 70/100

> Based on primary research, the company has not disclosed annual revenue and growth rate from battery fire safety products and services.

SUPPORTING POINTS

- Not available

Geographic Footprint & Channel Network — scored 76/100

> The evaluation rests on the company's verified footprint being limited to its primary headquarters and assembly site in the Netherlands, alongside a single localized regional partnership in Europe. Based on primary research, the company has not disclosed the number of active global distributors, commercial sales volume, or permanent international office networks outside of Europe.

SUPPORTING POINTS

- The company reports its assembly, warehouse, and primary office are located at a single site in Heinenoord, Netherlands.
- Lithium Safety Containers B.V. (operating under LithiumSafe) announced an exclusive distribution partnership in the D-A-CH region with LogBATT GmbH in December 2025.

Installed Base & Market Share — scored 98/100

> The rating rests on the verified fleetwide integration of the company's containment systems by a major commercial aviation group alongside self-reported deployment figures exceeding 10,000 units. Based on primary research, the company has not disclosed its precise overall market share percentage or a detailed breakdown of its installed base across industrial and marine applications.

SUPPORTING POINTS

- The company reports that more than 10,000 units are deployed worldwide.
- We verified that LithiumSafe Fire Containment Kits are installed fleetwide across the Lufthansa Group.

Funding & Strategic Partnerships — scored 73/100

> We verified that the brand operates as an established provider of battery fire containment equipment, though it lacks public disclosure of institutional funding rounds or equity-backed strategic partnerships. Based on primary research, the company has not disclosed any venture funding rounds, external investors, or formalized joint ventures.

SUPPORTING POINTS

- The company is a brand/trademark operated by Forfyre B.V., based in the Netherlands, which originally developed its containment bags at the request of a major European aviation company.

TL-X ● Evolving Players X 88 Y 70 Overall 79

X — PRODUCT CAPABILITY

Fire Suppression Technology & Performance — scored 93/100

> We verified that the company's capabilities rest on independent live-fire validations confirming successful suppression across multiple battery module sizes up to 10 kWh and KFI type approval for portable systems. Based on primary research, the company has not disclosed comprehensive third-party performance metrics regarding specific temperature reduction numbers or formal UL 9540A certification data.

SUPPORTING POINTS

- The company reports a gas-safety authority ESS live-fire test where its fixed suppression agent (FS PAD) confirmed complete suppression on 1, 2.5, and 10 kWh modules with no re-ignition within 10 minutes.
- The company's XMOR 119 Extinguisher Model TL-XBF 3.3 holds a KFI type approval for controlling flammable gases, cooling through penetration, and halting thermal-runaway propagation on a 1.2kWh lithium battery fire.

Battery Chemistry Compatibility & Coverage — scored 90/100

> We verified that the company's capability rests on its KFI-approved thermal runaway suppression agents tested up to 10 kWh modules. Based on primary research, the company has not disclosed a compatibility list or detailed test data outlining its performance across specific battery chemistries, such as NMC and LFP, or varied form factors including cylindrical, prismatic, and pouch cells.

SUPPORTING POINTS

- The company reports that its XMOR 119 suppression agent and fixed systems have achieved Korea Fire Institute (KFI) type approval for suppressing small lithium-ion battery fires, confirmed via live-fire module testing on energy storage system (ESS) modules ranging from 1 kWh to 10 kWh.

Product Portfolio Breadth & Modularity — scored 89/100

> We verified that the company's product range spans from localized in-pack suppression pads up to 50L module/pack units and vehicle spray systems across diverse application verticals. Based on primary research, the company has not disclosed granular cell-level sensor integration details or modular customization options in its public catalogs.

SUPPORTING POINTS

- The company reports a fire safety lineup including the XMOR 119 portable extinguisher (3.3L capacity certified for 1.2kWh lithium battery fires), the XMOR 119 Agent (200L spray/sprinkler system variant), the FS PAD suppression pad for early in-pack containment, and a 50L suppression unit.
- The company reports scalable implementations across diverse form factors targeting distinct customer applications, including energy storage systems (ESS/BESS), electric vehicles (EV), UPS packs, and data centers.

R&D and Innovation Pipeline — scored 75/100

> The assessment rests on the company's timeline of patent registrations and initial research milestones dating back to 2016. Based on primary research, the company has not disclosed audited R&D financial expenditures or independent technical evaluations of its core fire suppression technologies.

SUPPORTING POINTS

- The company reports registering domestic patents for a fire suppressant and a suppression sheet in 2024, following its establishment and the construction of a battery-specific extinguishing agent production line.
- The company reports conducting lithium-ion thermal runaway control research and developing advanced extinguishing technology starting in 2018, expanding upon electrical fire safety and material development research initiated in 2016.

Certifications & Standards Compliance — scored 92/100

> We verified that the company holds domestic KFI type approval and has completed initial NTA 8133 safety testing for its battery fire extinguishers and suppression agents. Based on primary research, the company has not disclosed compliance certificates or test reports for prominent international battery safety standards like UL 9540A, NFPA 855, or IEC 62619.

SUPPORTING POINTS

- The company reports that its XMOR 119 Extinguisher (Model TL-XBF 3.3) and XMOR 119 extinguishing agent hold KFI type approval (Class A and K) and performance certification for small lithium-ion battery fire suppression from South Korea.
- The company reports that its fire safety systems have undergone overseas safety testing under NTA 8133, with the certificate currently listed as being issued.

Y — BUSINESS CAPABILITY

Named Customers & Contracts — scored 65/100

> We verified that the company highlights various technical product specifications and standard applications for energy storage systems and electric vehicles on its official platform. Based on primary research, the company has not disclosed named customers, contract values, or specific partnership announcements.

SUPPORTING POINTS

- Not available

Revenue & Growth — scored 73/100

> Based on primary research, the company has not disclosed its annual revenue or year-over-year growth percentages from battery fire safety products and services.

SUPPORTING POINTS

- Not available

Geographic Footprint & Channel Network — scored 76/100

> The assessment rests on the verified single-country operational presence of the manufacturer in South Korea, alongside participation in European trade exhibitions to promote its fire mitigation product line. Based on primary research, the company has not disclosed its formal international distributor footprint, the total count of operational regional sales offices, or dedicated channel network partners outside of its domestic base.

SUPPORTING POINTS

- YookSong Co., Ltd., the manufacturer of the TL-X battery fire safety system, operates its primary corporate headquarters and office in Anseong-si, Gyeonggi-do, South Korea.
- YookSong showcased the TL-X battery fire safety system internationally at the INTERSCHUTZ 2026 trade show in Hannover, Germany.

Installed Base & Market Share — scored 69/100

> We verified that the company manufactures the XMOR 119 line of battery fire extinguishers and suppression pads. Based on primary research, the company has not disclosed its field-deployed unit counts or percentage share of the battery fire safety market.

SUPPORTING POINTS

- Not available

Funding & Strategic Partnerships — scored 65/100

> We verified that no specific public funding rounds, investor names, joint ventures, or strategic alliances with battery manufacturers have been confirmed for the company. Based on primary research, the company has not disclosed any investment raised or official stakeholder partnerships.

SUPPORTING POINTS

- Not available

EV Fire Solutions Pty Ltd EV Fire Solutions ● Evolving Players X 80 Y 76 Overall 78

X — PRODUCT CAPABILITY

Fire Suppression Technology & Performance — scored 87/100

> The rating rests on the company's baseline testing compliance under European EN standards for its commercial blankets and extinguishers. Based on primary research, the company has not disclosed independent test results under comprehensive lithium-ion fire protocols such as UL 9540A, specific suppression time figures, or precise temperature reduction metrics.

SUPPORTING POINTS

- The company reports that its fire containment blankets are tested to European EN 13501 fire safety standards, and its portable lithium-ion fire extinguishers comply with international standards EN-3-7-2004 and A1-2007.
- The company's silicone-coated lithium-ion battery fire containment blankets have a stated maximum working temperature of 1100°C to control thermal runaway events.

Battery Chemistry Compatibility & Coverage — scored 75/100

> We verified that the company offers specialised fire protection products such as blankets and extinguishers for lithium-ion battery and electric vehicle applications. Based on primary research, the company has not disclosed specific compatibility lists, test data across different chemistries (such as NMC or LFP), or application notes addressing battery cell form factors.

SUPPORTING POINTS

- Not available

Product Portfolio Breadth & Modularity — scored 83/100

> The rating rests on the company's verified portfolio of portable fire extinguishers, custom-built containment bags, and high-temperature fire blankets designed to mitigate thermal runaway in light electric and commercial vehicles. Based on primary research, the company has not disclosed cell-level or pack-integrated sensors, built-in fixed modular suppression systems, or scalable module-level containment options.

SUPPORTING POINTS

- The company reports offering a product catalog that includes EV fire blankets rated to 1700°C, portable lithium-ion fire extinguishers (1L, 4L, and 9L variants), fire curtains, and fire-resistant lithium-ion battery containment bags.
- The company reports manufacturing custom containment bags and fire blankets tailored to specific dimensions and design requirements for battery transit, storage, and quarantine.

R&D and Innovation Pipeline — scored 65/100

> The rating rests on the company's status as a specialized distributor and custom configurator of externally tested European-standard safety equipment rather than a developer of next-generation proprietary technologies. Based on primary research, the company has not disclosed R&D spending figures, active internal engineering pipelines, or patented technologies under its own name.

SUPPORTING POINTS

- The company offers a range of fire protection products, including EV fire blankets independently tested to European standard EN 13501-1:2018 and lithium-ion fire extinguishers certified to EN 3-7:2004+A1:2007.
- The company reports that its product lines—such as custom-dimension containment bags and high-silica blankets—are backed by industry collaboration, though it relies on sourcing and supplying rather than proprietary core hardware development.

Certifications & Standards Compliance — scored 90/100

> The rating rests on the company's compliance with European fire protection standards for its handheld extinguishers and fire blankets, as there is currently no specific Australian standard for lithium-ion fire safety. Based on primary research, the company has not disclosed compliance or test reports for battery-specific standards such as UL 9540A, NFPA 855, or IEC 62619.

SUPPORTING POINTS

- The company reports that its lithium-ion fire extinguishers comply with European and international standards EN 3-7:2004+A1:2007.
- The company reports that its heavy-duty vehicle fire blanket is classified A2-s1, d0 to EN 13501-1:2018, independently tested by SGS.

Y — BUSINESS CAPABILITY

Named Customers & Contracts — scored 87/100

> The evaluation rests on the company's active presence as an approved New South Wales government procurement supplier and its reported base of over 400 protected entities across commercial and municipal sectors. Based on primary research, the company has not disclosed specific individual contract values or formalized partnership announcements with original equipment manufacturers (OEMs) or battery producers.

SUPPORTING POINTS

- The company reports serving over 400 protected customers, including local councils, state government agencies, hospitals, aged care facilities, fleet operators, and strata buildings across Australia.
- The company is listed as an approved supplier on the New South Wales government procurement platform (buy.nsw) for fire safety equipment procurement by state agencies and councils.

Revenue & Growth — scored 70/100

> Based on primary research, the company has not disclosed its annual revenue or year-over-year growth rates.

SUPPORTING POINTS

- Not available

Geographic Footprint & Channel Network — scored 76/100

> We verified that the company's confirmed operations and distribution network are concentrated in Australia, leaning heavily on domestic online retail and public procurement channels. Based on primary research, the company has not disclosed the number of international offices, physical global distributors, or specific overseas sales partners it maintains.

SUPPORTING POINTS

- The company reports serving over 400 protected customers and supplying local councils, state government agencies, hospitals, aged care facilities, fleet operators, and strata buildings across Australia.
- The company holds active registration with a main business location in New South Wales (NSW), Australia, and distributes its specialized battery fire blankets and containment bags online through Bunnings Marketplace.

Installed Base & Market Share — scored 66/100

> Based on primary research, the company has not disclosed its installed unit counts or verified market share metrics.

SUPPORTING POINTS

- Not available

Funding & Strategic Partnerships — scored 81/100

> The rating rests on the company's self-reported general partnerships with safety innovators and a community sponsorship agreement. Based on primary research, the company has not disclosed any venture funding rounds, institutional investors, or joint ventures with battery manufacturers.

SUPPORTING POINTS

- The company reports it has partnered with leading safety innovators to offer specialized fire safety equipment.
- The company announced it is a sponsor of the ANU Solar Racing Team to support safety and innovation on the solar racing circuit.

LioShield Safety Systems - FZCO LioShield ● Evolving Players X 82 Y 66 Overall 74

X — PRODUCT CAPABILITY

Fire Suppression Technology & Performance — scored 87/100

> The evaluation rests on the company's self-reported technical specifications for its multi-layer thermal containment boxes and its ongoing validation process. Based on primary research, the company has not disclosed independent third-party performance metrics, verified suppression times, or standardized test results like UL 9540A.

SUPPORTING POINTS

- The company reports that its PyroSafe containment units withstand heat up to 1200 °C, feature a multi-layer thermal barrier system, incorporate an emergency venting system for controlled gas release, and utilize CoreSense Smart Monitoring sensors for local visual and audible temperature alerts.
- The company reports that its products are under TÜV testing for international safety and performance standards, with certification expected by December 2025.

Battery Chemistry Compatibility & Coverage — scored 78/100

> We verified that the company highlights its thermal containment units for a wide array of marine and industrial personal electric mobility devices and tools, but no technical matrices detailing compatibility with specific chemical compositions or battery form factors are provided. Based on primary research, the company has not disclosed comprehensive compatibility lists, test data across distinct chemistries (such as NMC or LFP), or application notes defining form factor boundaries.

SUPPORTING POINTS

- The company reports that its PyroSafe units are engineered to isolate and control lithium battery fires from onboard devices, including drones, tools, e-boards, e-foils, sea scooters, and removable e-propulsion packs.

Product Portfolio Breadth & Modularity — scored 83/100

> The rating rests on the company's confirmed modular product line of fire safety boxes integrated with temperature monitoring sensors and pressure relief systems for small-to-medium industrial and marine battery containment. Based on primary research, the company has not disclosed scalability evidence or a product catalog expanding to automated or modular pack-level suppression systems for large grid-scale energy storage or electric vehicle platforms.

SUPPORTING POINTS

- The company reports its PyroSafe line of modular storage boxes (including S1, M1, and L1 models) that use a multi-layer thermal containment barrier certified to withstand 1200 °C temperatures alongside a CoreSense smart monitoring system consisting of PyroBridge and PyroNode sensors.
- The company reports its containment boxes feature an emergency venting system built with controlled venting channels designed to release pressure and gases during thermal runaway while sealing flames inside.

R&D and Innovation Pipeline — scored 75/100

> The assessment rests on the company's self-reported technical specifications for its PyroSafe containment product line, including built-in thermal barrier and smart monitoring components. Based on primary research, the company has not disclosed its specific R&D expenditure or registered patent filings to verify its innovation pipeline through independent sources.

SUPPORTING POINTS

- The company reports the development of its PyroSafe thermal containment boxes, which utilize a proprietary multi-layer thermal barrier, an aramid-carbon composite core, an emergency venting system, and CoreSense smart monitoring.
- The company reports that its products are currently under TÜV testing for international safety and performance standards, with certification expected by December 2025.

Certifications & Standards Compliance — scored 88/100

> The rating rests on the fact that LioShield Safety Systems - FZCO has only self-reported an in-progress safety testing status rather than completed third-party compliance. Based on primary research, the company has not disclosed completed certifications or verified test listings for standards like UL 9540A, NFPA 855, or IEC 62619.

SUPPORTING POINTS

- The company reports its products are currently under TÜV testing for international safety and performance standards, with certification expected by December 2025.

Y — BUSINESS CAPABILITY

Named Customers & Contracts — scored 65/100

> Based on primary research, the company has not disclosed named customers, contract values, or public partnership announcements within the electric vehicle, marine, or energy storage sectors.

SUPPORTING POINTS

- Not available

Revenue & Growth — scored 67/100

> Based on primary research, the company has not disclosed annual revenue and growth rate from battery fire safety products and services.

SUPPORTING POINTS

- Not available

Geographic Footprint & Channel Network — scored 65/100

> The assessment rests on the company's confirmed physical presence and registration in Dubai, UAE, which serves as its primary operational footprint. Based on primary research, the company has not disclosed the number of countries served, additional international offices, or specific third-party distributors and sales partners within its channel network.

SUPPORTING POINTS

- The company maintains an office location registered as Lioshield Safety Systems – FZCO at the IFZA Business Park, Building A2, Dubai Silicon Oasis Industrial Area, Dubai, UAE.

Installed Base & Market Share — scored 66/100

> Based on primary research, the company has not disclosed its installed base or market share data.

SUPPORTING POINTS

- Not available

Funding & Strategic Partnerships — scored 67/100

> Based on primary research, the company has not disclosed its investment history or alliance agreements.

SUPPORTING POINTS

- Not available

Lion Battery ● Evolving Players

X 67 Y 75 Overall 71

X — PRODUCT CAPABILITY

Fire Suppression Technology & Performance — scored 65/100

> Based on primary research, the company has not disclosed fire suppression technologies, thermal management test results, or official safety certifications such as UL 9540A.

SUPPORTING POINTS

- Not available

Battery Chemistry Compatibility & Coverage — scored 65/100

> Based on primary research, the company has not disclosed any technical documentation, application notes, or test logs demonstrating fire containment or suppression capabilities for lithium-ion variations.

SUPPORTING POINTS

- Not available

Product Portfolio Breadth & Modularity — scored 65/100

> The rating rests on the company's verified manufacture of fire-resistant storage enclosures for containing thermal runaways, though its core catalog consists of custom battery packs and basic protection accessories rather than dedicated cell-level or modular suppression systems. Based on primary research, the company has not disclosed a dedicated modular fire safety product suite mapping from cell-level sensors to module and pack-level active suppression systems.

SUPPORTING POINTS

- The company reports offering fire-resistant metal lithium battery safety boxes designed for the safe storage, charging, quarantine, handling, and transportation of lithium-ion, $LiFePO_4$, and LiPo batteries.
- The company reports supplying these safety boxes in both standard and custom sizes to meet the requirements of OEMs, EV companies, drone manufacturers, and industrial customers.

R&D and Innovation Pipeline — scored 75/100

> We verified that the firm's fire safety capabilities rest on self-reported entries for basic thermal containment enclosures and standard battery management system safety features. Based on primary research, the company has not disclosed R&D spending, active patent filings, advanced detection algorithms, or novel suppression agents.

SUPPORTING POINTS

- The company reports offering a range of fire-resistant metal battery safety boxes designed with optional pressure relief and vent designs for safe charging, storage, and transport of lithium-ion and $LiFePO_4$ batteries.
- The company reports engineering its battery management systems to manage Li-Ion, $LiFePO_4$, and NMC packs, providing real-time thermal protection against temperature extremes.

Certifications & Standards Compliance — scored 65/100

> Based on primary research, the company has not disclosed any fire safety certifications, test reports, or official listings with third-party verification bodies.

SUPPORTING POINTS

- Not available

Y — BUSINESS CAPABILITY

Named Customers & Contracts — scored 86/100

> The evaluation rests on the company's self-reported role as a manufacturer and global exporter of custom and standard-sized fire-resistant battery safety boxes designed to prevent risks associated with battery storage. Based on primary research, the company has not disclosed the names of specific customers, contract values, or verified commercial partnership agreements.

SUPPORTING POINTS

- The company reports manufacturing fire-resistant metal safety boxes for the safe storage, charging, transportation, and handling of lithium-ion and $LiFePO_4$ batteries, supplying them to OEMs, battery manufacturers, EV companies, drone manufacturers, defense organizations, and industrial customers.

Revenue & Growth — scored 65/100

> We verified that no specific revenue or financial disclosures exist for Lion Battery Technologies within the defined battery fire safety industry, as the joint venture is focused on platinum- and palladium-based electrode research for energy density optimization. Based on primary research, the company has not disclosed annual revenue and growth rate from battery fire safety products and services.

SUPPORTING POINTS

- Not available

Geographic Footprint & Channel Network — scored 85/100

> The evaluation is based on the company's documented operational footprint inside India and its self-reported capacity to export fire-resistant containment boxes internationally. Based on primary research, the company has not disclosed verified quantitative evidence regarding its active distribution partners or the exact number of countries it currently serves.

SUPPORTING POINTS

- The company reports operating a warehouse location in Umargam, Valsad, Gujarat, India, and head office locations in Dahisar (Mumbai) and Vasai (Palghar), Maharashtra, India.
- The company reports supplying fire-resistant battery safety boxes to global distributors and offers export-ready packaging for customers worldwide.

Installed Base & Market Share — scored 71/100

> Assessment rests on the company's self-disclosed product lines for fire-resistant metal safety enclosures and battery management units with basic thermal protection. Based on primary research, the company has not disclosed installed unit counts, market share data, or third-party verified shipment numbers.

SUPPORTING POINTS

- The company reports manufacturing metal battery safety boxes and battery management systems with thermal protection from its facility in Umargam, Gujarat, India.

Funding & Strategic Partnerships — scored 67/100

> Based on primary research, the company has not disclosed its funding rounds or strategic partnerships.

SUPPORTING POINTS

- Not available

Parameter Definitions

X — Product Capability

1. **Fire Suppression Technology & Performance**
   The effectiveness and speed of fire suppression systems for battery fires, including detection, suppression agents, and thermal management. Evidence includes test results (e.g., UL 9540A), suppression time, and temperature reduction metrics.
2. **Battery Chemistry Compatibility & Coverage**
   The ability of fire safety products to handle various lithium-ion battery chemistries (NMC, LFP, etc.) and form factors (cylindrical, prismatic, pouch). Evidence includes compatibility lists, test data across chemistries, and application notes.
3. **Product Portfolio Breadth & Modularity**
   The range of fire safety products offered, from cell-level sensors to module and pack-level suppression systems, and their scalability for different applications. Evidence includes product catalogs, modular designs, and customization options.
4. **R&D and Innovation Pipeline**
   Investment in research and development for new fire safety technologies, such as advanced detection algorithms, novel suppression agents, and integration with battery management systems. Evidence includes patents, R&D spending, and new product launches.
5. **Certifications & Standards Compliance**
   Compliance with industry standards and certifications for battery fire safety, such as UL 9540A, NFPA 855, and IEC 62619. Evidence includes certificates, test reports, and listings with certification bodies.

Y — Business Capability

1. **Named Customers & Contracts**
   Publicly disclosed customers and contracts with battery manufacturers, electric vehicle OEMs, energy storage integrators, and other end-users. Evidence includes customer names, contract values, and partnership announcements.
2. **Revenue & Growth**
   Annual revenue and growth rate from battery fire safety products and services. Evidence includes financial statements, revenue breakdowns, and year-over-year growth percentages.
3. **Geographic Footprint & Channel Network**
   Presence in key markets and distribution channels for battery fire safety solutions. Evidence includes number of countries served, offices, distributors, and sales partners.
4. **Installed Base & Market Share**
   The number of fire safety systems deployed in the field and the company's share of the battery fire safety market. Evidence includes installed unit counts, market share reports, and third-party analyses.
5. **Funding & Strategic Partnerships**
   Investment raised and strategic alliances with battery manufacturers, research institutions, or other stakeholders. Evidence includes funding rounds, investors, joint ventures, and collaboration agreements.

Research Methodology

This quadrant was compiled independently through structured secondary research, supplemented by primary research where available. It is not based on a survey or on vendor self-submission. The steps below describe how companies were identified, screened, evidenced and scored.

1 Market Definition

The market scope and the type of company that qualifies were defined first. Five parameters were then set for each axis, each with a written definition: the X axis measures core capability (products, services or technology) and the Y axis measures business capability.

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2 Secondary and Primary Research

Secondary research covered company websites, filings, trade directories, industry publications and news. Primary research drew on B2B marketplace listings, where available, to confirm company offerings. Coverage was extended region by region until no new qualifying companies emerged.

→

3 Screening

Each candidate was screened against the market's inclusion criteria and required commercial role. Distributors, resellers, consultancies, contract manufacturers and media names were excluded, and duplicates consolidated. Companies whose main business is an adjacent segment are listed but not placed on the chart.

→

4 Evidence Compilation

Every screened company was assessed on all ten parameters; for chart companies, evidence was recorded parameter by parameter. Claims resting only on a company's own materials are reported as company-reported, partner technology is credited to the partnership, and employee-review sites are not used as customer evidence.

→

5 Scoring

Each parameter is rated 0–100 against its written definition, based on the evidence available. Ratings are analyst assessments under this methodology, not independently audited figures. X and Y are the simple average of the five parameter ratings on each axis; overall is the average of X and Y.

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6 Normalization & Selection

X and Y scores are rescaled to a 65–100 range across all companies assessed, preserving their rank order. The chart shows an equal number of companies from each of the four quadrants, with boundaries set at the median. All other companies are listed in the table below the chart.

Scoring inputs: each axis is built from five market-specific parameters, defined in the Parameter Definitions section. All parameters are equally weighted.